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Exact Computation for the Normalised Power Prior

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Master's Thesis presented to the Graduate Program in Applied Mathematics and Data Science of the School of Applied Mathematics as a requirement to obtain the degree of Master in Applied Mathematics and Data Science.

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Statement of Authorship

I hereby declare that the thesis submitted is my own work. All direct or indirect sources used are acknowledged as references. I further declare that I have not submitted this thesis at any other institution in order to obtain a degree.

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I would like to thank [many people]

"So, nobody told me what to do, and there was no preconception of what to do."
From *Giorgio By Moroder* (2013) by Daft Punk

Abstract

In statistical modelling with historical data, a central challenge is how to effectively integrate prior evidence with current observations. This issue is particularly pressing in clinical trials and medical applications, where using historical data can improve both the efficiency and robustness of inference. A natural Bayesian strategy for this purpose is the use of informative priors. Among them, the normalised power prior has attracted significant interest, as it provides a principled way to control the influence of historical information. The method introduces a power parameter, $\eta \in [0, 1]$, that governs the degree of borrowing: $\eta = 0$ discards the past data while $\eta = 1$ fully incorporates it. This flexibility makes the framework highly appealing in practice. Despite its advantages, the normalised power prior induces a doubly intractable posterior: the prior itself contains a normalising constant $c_0(\eta)$ that depends on the power parameter η and is unavailable in closed form for most models of interest, rendering standard Metropolis–Hastings inapplicable. Existing approaches therefore rely on approximate inference methods that often lack theoretical guarantees and explicit convergence properties. To overcome these limitations, we propose exact Markov chain Monte Carlo algorithms that avoid direct evaluation of the normalising constant ratio. The key insight is a thermodynamic identity expressing the log-ratio of normalising constants as a tractable conditional expectation under the power prior. Exploiting this identity, we construct non-negative unbiased estimators of the normalising constant ratio via Monte Carlo integration, which we embed within an Exchange Metropolis–Hastings scheme and a pseudo-marginal framework. We evaluate the method on Gaussian models under varying degrees of compatibility between historical and current data. Results demonstrate that the algorithm accurately reflects data conflict by shrinking η when the datasets are incompatible, while recovering correct posterior behaviour when they are aligned. Comparisons with analytical baselines confirm that our method achieves reliable estimates of both model parameters and the power parameter, with efficiency suitable for practical use. This provides a rigorous, exact alternative to approximate approaches, offering stronger guarantees for applications such as adaptive clinical trial design.

Keywords: Bayesian Inference; Power Prior; Doubly Intractable Distributions; Exact MCMC; Historical Data Analysis

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1 Introduction

`Luiz:` Second-to-last thing to be written. Don't sweat it. Leave it to *Papa*

- Describe NPP and notation; motivation with historical data (MANY CITATIONS).
- The NPP as a doubly-intractable problem (the prior is intractable)
- Mention that we start with approximate methods and then move on to exact stuff

In statistical modelling with historical data, a central challenge is effectively integrating information from past studies with current data. This is particularly relevant in clinical trials and medical research, where leveraging historical data can improve the efficiency and robustness of statistical inferences. A natural approach in this setting is the use of *informative priors*, which incorporate prior knowledge into the analysis, thereby enhancing parameter estimation and decision-making (Spiegelhalter, 2004; Gelman et al., 2013; Ibrahim et al., 2015).

Among the various methods for constructing informative priors, *power priors* (Ibrahim et al., 2015) have gained significant attention. They allow for controlled borrowing of information from historical data by adjusting a power parameter, η, which typically ranges between 0 and 1. When $\eta = 0$, no historical information is incorporated, whereas $\eta = 1$ fully integrates the historical data into the current analysis. This flexibility makes power priors particularly appealing in adaptive clinical trial design and other settings where historical evidence is available but its relevance is uncertain.

However, when η is itself treated as unknown and assigned a prior distribution, the resulting posterior is *doubly intractable*: the normalised power prior contains a normalising constant $c_0(\eta)$ that depends on η and cannot be evaluated in closed form for general likelihoods, preventing direct use of Metropolis–Hastings. Current approaches rely on *approximate methods* (Carvalho; Ibrahim, 2021) that lack theoretical guarantees and explicit convergence bounds. This work addresses this gap by developing exact Markov chain Monte Carlo algorithms (Murray; Ghahramani; MacKay, 2012; Gonçalves; Łatuszyński; Roberts, 2017; Vats et al., 2021) that avoid evaluating the intractable normalising constant ratio altogether, providing a rigorous computational baseline for inference under the normalised power prior.

2 Background

2.1 Bayesian Inference

2.1.1 Principles

Bayesian inference provides a probabilistic framework for updating uncertainty about unknown quantities after observing data.

Let D denote the observed data, and let $\theta \in \Theta$ denote an unknown parameter defined on a measurable parameter space $(\Theta, \mathcal{F})$. The Bayesian model is specified by a likelihood function $L(D \mid \theta)$ and a prior distribution $\pi_0(\theta)$.

The likelihood describes the information about θ contained in the data, while the prior represents information available before observing D. Bayes' theorem combines these two sources of information through the posterior distribution

$$\pi(\theta \mid D) = \frac{L(D \mid \theta)\pi_0(\theta)}{\int_\Theta L(D \mid t)\pi_0(t)\,\mathrm{d}t}. \tag{2.1}$$

The denominator in (2.1) is the marginal likelihood, or evidence, and is given by

$$m(D) = \int_\Theta L(D \mid t)\pi_0(t)\,\mathrm{d}t. \tag{2.2}$$

When $m(D) < \infty$, the posterior distribution is well-defined and integrates to one. In this sense, Bayesian inference can be understood as a normalization procedure that transforms the unnormalized density $L(D \mid \theta)\pi_0(\theta)$ into a probability distribution over Θ.

For any posterior-integrable function $h : \Theta \to \mathbb{R}$, posterior summaries are often expressed through expectations of the form

$$\mathbb{E}_{\pi(\theta|D)}[h(\theta)] = \int_\Theta h(\theta)\pi(\theta \mid D)\,\mathrm{d}\theta. \tag{2.3}$$

Examples include posterior means, posterior variances, marginal probabilities, credible intervals, and posterior predictive quantities. The main computational challenge is that the posterior distribution is usually available only up to a normalizing constant.

Indeed, many Bayesian algorithms only require the unnormalized posterior density

$$\gamma(\theta) = L(D \mid \theta)\pi_0(\theta), \tag{2.4}$$

because the marginal likelihood $m(D)$ does not depend on θ.

However, the marginal likelihood is still important for model comparison, empirical Bayes procedures, and doubly intractable models. In low-dimensional conjugate models, the integral in (2.2) may be available analytically.

In modern Bayesian models, this integral is often unavailable, and posterior expectations such as (2.3) must be approximated numerically. This motivates the use of simulation-based methods.

2.1.2 Monte Carlo Methods

Monte Carlo methods approximate integrals by random sampling.

Let π be a probability distribution on $(\Theta, \mathcal{F})$, and suppose that the quantity of interest is

$$\mu = \mathbb{E}_\pi[h(\theta)] = \int_\Theta h(\theta)\pi(\theta)\,\mathrm{d}\theta. \tag{2.5}$$

If $\theta_1, \ldots, \theta_N$ are independent samples from π, then the Monte Carlo estimator of μ is

$$\widehat{\mu}_N = \frac{1}{N}\sum_{i=1}^{N} h(\theta_i). \tag{2.6}$$

Under standard integrability conditions, the law of large numbers gives

$$\widehat{\mu}_N \xrightarrow[N\to\infty]{a.s.} \mu. \tag{2.7}$$

If $h(\theta)$ has finite variance under π, the central limit theorem gives

$$\sqrt{N}(\widehat{\mu}_N - \mu) \xrightarrow[N\to\infty]{d} \mathcal{N}\big(0, \sigma_h^2\big), \tag{2.8}$$

where

$$\sigma_h^2 = \mathrm{Var}_\pi\,[h(\theta)]. \tag{2.9}$$

The important feature of the Monte Carlo approximation is that its convergence rate is typically $N^{-1/2}$ and does not directly depend on the dimension of Θ. This makes Monte Carlo methods attractive for Bayesian inference, where posterior distributions are often high-dimensional.

The difficulty is that independent sampling from $\pi(\theta \mid D)$ is rarely possible. Markov chain Monte Carlo methods address this difficulty by constructing a Markov chain whose invariant distribution is the desired posterior distribution.

The samples are then dependent, but posterior expectations can still be estimated using ergodic averages. If $(\theta_n)_{n\geq 0}$ is a Markov chain with invariant distribution π, then the MCMC estimator is

$$\widehat{\mu}_N^{\mathrm{MCMC}} = \frac{1}{N}\sum_{n=1}^{N} h(\theta_n). \tag{2.10}$$

Under suitable irreducibility, recurrence, and integrability conditions, the ergodic theorem gives

$$\widehat{\mu}_N^{\mathrm{MCMC}} \xrightarrow[N\to\infty]{a.s.} \mathbb{E}_\pi[h(\theta)]. \tag{2.11}$$

The price paid for replacing independent samples with a Markov chain is autocorrelation.

When samples are correlated, the effective amount of information in N iterations may be much smaller than N independent draws. This motivates the use of diagnostics such as effective sample size, Monte Carlo standard errors, trace plots, and comparisons across independent chains. Another important practical issue is initialization.

A Markov chain usually starts from a distribution different from π, and therefore the early iterations may not be representative of the target distribution. The usual practical response is to discard an initial portion of the chain, known as burn-in.

However, burn-in is an asymptotic correction rather than an exact one. This distinction is central to the later discussion of exact and unbiased MCMC methods.

2.2 Classical Markov Chain Monte Carlo Methods

Markov chain Monte Carlo methods construct a transition kernel P on $(\Theta, \mathcal{F})$ such that the posterior distribution π is invariant.

This means that, for every measurable set $A \in \mathcal{F}$,

$$\int_\Theta P(\theta, A)\pi(\theta)\,\mathrm{d}\theta = \pi(A). \tag{2.12}$$

Equivalently, if $\theta_n \sim \pi$ and $\theta_{n+1} \mid \theta_n \sim P(\theta_n, \cdot)$, then $\theta_{n+1} \sim \pi$. Invariance alone does not guarantee convergence from arbitrary starting points.

For convergence, the Markov chain must also satisfy appropriate irreducibility and recurrence conditions. A common sufficient condition used to establish invariance is detailed balance.

The transition kernel P satisfies detailed balance with respect to π if

$$\pi(\theta)P(\theta, \mathrm{d}\theta') = \pi(\theta')P(\theta', \mathrm{d}\theta) \tag{2.13}$$

as measures on $\Theta \times \Theta$.

Detailed balance implies invariance and gives a convenient route for proving that a Markov chain targets the desired posterior distribution. The two classical algorithms discussed in this section are Gibbs sampling and Metropolis-Hastings.

Gibbs sampling exploits conditional distributions that can be sampled exactly. Metropolis-Hastings is more general and only requires evaluation of the target density up to a normalizing constant.

Both algorithms are asymptotically exact in the sense that their ergodic averages converge to posterior expectations under suitable conditions. They are not exact in the finite-time sense, because samples obtained after a finite number of iterations still depend on the initial state.

2.2.1 Gibbs Sampling

Gibbs sampling is designed for posterior distributions whose full conditional distributions are available and easy to sample from.

Suppose that the parameter can be partitioned into K blocks,

$$\theta = (\theta_1, \ldots, \theta_K), \qquad \Theta = \Theta_1 \times \cdots \times \Theta_K. \tag{2.14}$$

For each block θ_k, define the complementary vector

$$\theta_{-k} = (\theta_1, \ldots, \theta_{k-1}, \theta_{k+1}, \ldots, \theta_K). \tag{2.15}$$

The full conditional distribution of the k-th block is

$$\pi(\theta_k \mid \theta_{-k}, D). \tag{2.16}$$

A systematic-scan Gibbs sampler updates the blocks sequentially. Starting from $\theta^{(n)} = \left(\theta_1^{(n)}, \ldots, \theta_K^{(n)}\right)$, one iteration produces $\theta^{(n+1)}$ by sampling

$$\theta_1^{(n+1)} \sim \pi\left(\theta_1 \mid \theta_2^{(n)}, \ldots, \theta_K^{(n)}, D\right), \tag{2.17}$$

$$\theta_2^{(n+1)} \sim \pi\left(\theta_2 \mid \theta_1^{(n+1)}, \theta_3^{(n)}, \ldots, \theta_K^{(n)}, D\right), \tag{2.18}$$

$$\vdots$$

$$\theta_K^{(n+1)} \sim \pi\left(\theta_K \mid \theta_1^{(n+1)}, \ldots, \theta_{K-1}^{(n+1)}, D\right). \tag{2.19}$$

Each update leaves the posterior distribution invariant because it samples exactly from a conditional distribution of the target. The composition of these invariant updates also leaves the posterior distribution invariant.

The main advantage of Gibbs sampling is that every proposed block update is accepted automatically. In this sense, the algorithm avoids the tuning of an acceptance probability.

On the other hand, the most relevant limitation is that the required full conditional distributions may not be available in closed form. Even when they are available, strong posterior dependence between blocks can lead to slow mixing.

This is especially relevant in hierarchical models, latent variable models, and models with highly correlated parameters. In such cases, blocking, reparameterization, or hybrid Gibbs-within-Metropolis strategies may be necessary.

2.2.2 Metropolis-Hastings

The Metropolis-Hastings algorithm constructs a Markov chain using a proposal distribution and an accept-reject correction. Let π denote the target posterior density, known up to a normalizing constant.

Let $q(\theta' \mid \theta)$ be a proposal density for moving from the current state θ to a proposed state θ'. Given the current state θ, the algorithm first samples

$$\theta' \sim q(\cdot \mid \theta). \tag{2.20}$$

The proposal is then accepted with probability

$$\alpha(\theta, \theta') = \min\left\{1, \frac{\pi(\theta')q(\theta \mid \theta')}{\pi(\theta)q(\theta' \mid \theta)}\right\}. \tag{2.21}$$

If the proposal is accepted, the next state is θ'. If the proposal is rejected, the chain remains at θ. Thus,

$$\theta_{n+1} = \begin{cases} \theta', & \text{with probability } \alpha(\theta_n, \theta'), \\ \theta_n, & \text{otherwise.} \end{cases} \tag{2.22}$$

The acceptance probability in (2.21) depends only on ratios of the target density. Therefore, if $\pi(\theta) = \gamma(\theta)/Z$ for an unknown normalizing constant Z, then Z cancels from the ratio.

This cancellation is one of the main reasons why Metropolis-Hastings is useful in Bayesian computation. The Metropolis-Hastings transition kernel satisfies detailed balance with respect to π.

Indeed, the accept-reject correction symmetrizes the probability flow between θ and θ'. Consequently, π is invariant for the Markov chain.

A common special case is the random-walk Metropolis algorithm. In this case,

$$q(\theta' \mid \theta) = q(\theta - \theta'), \tag{2.23}$$

so that $q(\theta' \mid \theta) = q(\theta \mid \theta')$. The acceptance probability then reduces to

$$\alpha(\theta, \theta') = \min\left\{1, \frac{\pi(\theta')}{\pi(\theta)}\right\}. \tag{2.24}$$

The efficiency of Metropolis-Hastings depends strongly on the proposal distribution. If the proposal variance is too small, the chain accepts many proposals but explores the state space slowly.

If the proposal variance is too large, the chain proposes distant states that are often rejected. This trade-off is one of the central practical issues in MCMC.

Metropolis-Hastings is asymptotically exact under appropriate conditions, but it does not remove the finite-time dependence on the initial state. For this reason, convergence diagnostics and burn-in choices remain important in practical applications. In doubly intractable models, even the ratio in (2.21) may contain unknown normalizing constants that depend on θ.

This is one of the settings where standard Metropolis-Hastings is no longer directly applicable and where the methods studied in this dissertation become necessary.

2.3 Doubly Intractable Distributions

Doubly intractable distributions arise in Bayesian inference when the likelihood takes the form

$$\pi(\theta \mid x) \propto p(\theta)\frac{h(x \mid \theta)}{Z(\theta)}, \tag{2.25}$$

where the normalizing constant $Z(\theta)$ depends on the parameter θ and cannot be evaluated pointwise.

This setting is common in models such as exponential random graph models, Markov point processes, and, more generally, in likelihoods derived from Gibbs measures. In the context of Normalized Power Priors (NPP), this issue naturally appears due to the presence of parameter-dependent normalizing constants, making standard Bayesian computation challenging.

2.4 The Normalised Power Prior

Adame: Write some motivation here.

2.4.1 Formulation

Recalling the framework of Section 2.1, the posterior distribution of θ given data D takes the form of (2.1), with $\pi_0(\theta)$ serving as the initial prior. The normalised power prior framework replaces this initial prior by a data-informed version constructed from historical data $D_0 = \{d_{01}, \ldots, d_{0N_0}\}$.

Definition 2.4.1 (Power prior). *Let $D_0 = \{d_{01}, d_{02}, \ldots, d_{0N_0}\}$ with $d_{0i} \in \mathcal{X}^p$ be the historical data, and let $L_0(D_0 \mid \theta)$ be a likelihood function that is finite for all $\theta \in \Theta \subseteq \mathbb{R}^p$. Then, the power prior is defined as*

$$\pi_\eta(\theta \mid D_0) \propto L_0(D_0 \mid \theta)^\eta \pi_0(\theta), \tag{2.26}$$

where $\pi_0(\theta)$ is the initial prior distribution of θ and η is the (fixed) power parameter.

To accommodate uncertainty in the choice of η, we treat it as a random variable and assign a prior distribution $\pi_A(\eta \mid \varphi)$, indexed by hyperparameters φ.

The unnormalised joint power prior over (θ, η) is then

$$\pi^\star(\theta, \eta \mid D_0, \varphi) \propto L_0(D_0 \mid \theta)^\eta \, \pi_0(\theta) \, \pi_A(\eta \mid \varphi). \tag{2.27}$$

To normalise this joint prior, one requires the normalising constant

$$c_0(\eta) = \int_\Theta L_0(D_0 \mid t)^\eta \, \mathrm{d}P_0(t), \tag{2.28}$$

which is unavailable in closed form for general likelihoods.

Given new data D, the marginal posterior distribution of η is

$$\pi(\eta \mid D, D_0, \varphi) \propto \frac{\pi_A(\eta \mid \varphi)}{c_0(\eta)} \int_\Theta L(D \mid t) \, L_0(D_0 \mid t)^\eta \, \mathrm{d}P_0(t). \tag{2.29}$$

Since neither $c_0(\eta)$ nor the integral in (2.29) is tractable in general, this is a *doubly intractable* posterior in the sense of Section 2.3. The computational challenge this poses, and the strategies developed to address it, are the central subject of this thesis.

2.4.2 The derivative of the normalising constant

Under regularity conditions permitting differentiation under the integral sign, the thermodynamic identity (proved in Appendix A.1) gives

$$\frac{\mathrm{d} \log c_0(\eta)}{\mathrm{d}\eta} = \mathbb{E}_{\pi_\eta}\Big[\log L_0(D_0 \mid \theta)\Big], \tag{2.30}$$

where $\mathbb{E}_{\pi_\eta}[\cdot]$ denotes expectation under the power prior $\pi_\eta(\theta \mid D_0)$.

This identity plays a central role in the exact inference methods of Chapter 4: it expresses the log-derivative of an intractable normalising constant as a tractable posterior expectation, which can be estimated by Monte Carlo simulation.

2.4.3 Other Historical Data Modelling Approaches

The normalised power prior is one among several principled Bayesian approaches to incorporating historical information. The *commensurate prior* of (Hobbs et al., 2011) models the relationship between historical and current parameters through a shared hierarchical structure, allowing the degree of borrowing to adapt to the compatibility between studies.

Meta-analytic-predictive (MAP) priors (Neuenschwander et al., 2010) derive an informative prior for the current study by fitting a hierarchical model to a collection of historical studies and marginalising over the between-study heterogeneity.

Robust mixture priors (Schmidli et al., 2014) address the risk of prior-data conflict by mixing an informative component derived from historical evidence with a vague component, ensuring that the prior retains a non-negligible weight in regions not supported by the historical data.

Each of these alternatives makes different modelling assumptions about the exchangeability of the historical and current parameters, and each leads to different computational challenges. The normalised power prior is particularly appealing when a single scalar parameter $\eta \in [0, 1]$ suffices to capture the degree of historical relevance, and when the historical and current data share the same likelihood structure.

2.5 MCMC Methods for Intractable Normalising Constants

Many Bayesian models involve posterior distributions whose density can only be evaluated up to a normalising constant. This is not an obstacle for ordinary Metropolis–Hastings when the unknown normalising constant does not depend on the parameter being updated. However, the situation is considerably harder when the normalising constant itself depends on the parameter, as is the case for the normalised power prior.

This setting is usually called a doubly intractable problem, because the posterior distribution contains an intractable normalizing constant inside the likelihood or prior structure itself. The normalized power prior is an example of this phenomenon, since the normalizing constant $c_0(\eta)$ depends on the discount parameter η.

In this section, we review three families of methods that are relevant for this thesis. First, we discuss the exchange algorithm, which removes the intractable ratio of normalizing constants by introducing an auxiliary draw from a proposal-dependent distribution. Second, we discuss Barker's algorithm and Bernoulli factory methods, which replace the explicit evaluation of an acceptance probability by an exact randomization procedure. Third, we discuss divide-and-conquer Bernoulli factories, which aim to reduce the computational cost of Bernoulli factory methods in models with many likelihood factors.

2.5.1 The Exchange Algorithm

Consider a family of unnormalized densities $f_\eta(\theta)$ with normalizing constant

$$c(\eta) = \int_\Theta f_\eta(\theta)\mathrm{d}\theta. \tag{2.31}$$

Suppose that the target distribution has density

$$\pi(\eta \mid y) \propto \frac{\gamma(\eta \mid y)}{c(\eta)}. \tag{2.32}$$

A standard Metropolis–Hastings proposal, as discussed in section 2.2.2, from η to η' requires the ratio

$$\frac{\pi(\eta' \mid y)}{\pi(\eta \mid y)} = \frac{\gamma(\eta' \mid y)}{\gamma(\eta \mid y)}\frac{c(\eta)}{c(\eta')}. \tag{2.33}$$

The ratio $c(\eta)/c(\eta')$ is generally unavailable. This prevents the direct application of ordinary Metropolis–Hastings.

The exchange algorithm of Murray, Ghahramani and MacKay (2012) introduces an auxiliary variable $x^\star$ sampled exactly from the normalized model at the proposed value η'. That is,

$$x^\star \sim \frac{f_{\eta'}(x)}{c(\eta')}. \tag{2.34}$$

The Metropolis–Hastings ratio is then constructed on an augmented space so that the unknown normalizing constants cancel. In its simplest form, the exchange ratio is

$$R_{\mathrm{ex}}(\eta, \eta'; x^\star) = \frac{\gamma(\eta' \mid y)}{\gamma(\eta \mid y)}\frac{q(\eta \mid \eta')}{q(\eta' \mid \eta)}\frac{f_\eta(x^\star)}{f_{\eta'}(x^\star)}. \tag{2.35}$$

The corresponding acceptance probability is

$$\alpha_{\mathrm{ex}}(\eta, \eta') = \min\left\{1, R_{\mathrm{ex}}(\eta, \eta'; x^\star)\right\}. \tag{2.36}$$

The key point is that R_{ex} contains only unnormalized densities and proposal terms. Therefore, the intractable ratio $c(\eta)/c(\eta')$ is not evaluated directly.

In this thesis, the exchange algorithm is used as a benchmark exact method whenever exact simulation from the powered historical posterior is available. This provides a clean reference point for comparing approximate estimators and Bernoulli factory constructions.

2.5.2 Barker's Algorithm for Exact Inference

Barker's algorithm (Barker, 1965) is an alternative accept–reject rule for Markov chain Monte Carlo. For a target distribution $\pi(\theta)$ and proposal density $q(\theta' \mid \theta)$, define the usual Metropolis–Hastings odds ratio as

$$R(\theta, \theta') = \frac{\pi(\theta')q(\theta \mid \theta')}{\pi(\theta)q(\theta' \mid \theta)}. \tag{2.37}$$

The Barker acceptance probability is

$$\alpha_B(\theta, \theta') = \frac{R(\theta, \theta')}{1 + R(\theta, \theta')}. \tag{2.38}$$

Equivalently,

$$\alpha_B(\theta, \theta') = \frac{\pi(\theta')q(\theta \mid \theta')}{\pi(\theta)q(\theta' \mid \theta) + \pi(\theta')q(\theta \mid \theta')}. \tag{2.39}$$

This acceptance probability satisfies detailed balance because

$$\frac{\alpha_B(\theta, \theta')}{\alpha_B(\theta', \theta)} = R(\theta, \theta'). \tag{2.40}$$

Therefore, Barker's algorithm leaves the target distribution invariant.

Barker's algorithm is usually less efficient than Metropolis–Hastings in the Peskun ordering. However, its acceptance probability has a useful algebraic form. In particular, it can be implemented by Bernoulli factory methods when the odds ratio can be represented through probabilities that can be simulated but not evaluated. This makes Barker's algorithm useful for some intractable likelihood problems (Gonçalves; Łatuszyński; Roberts, 2017; Vats et al., 2021).

The important distinction is that Barker's algorithm is not required for the exchange algorithm. The exchange algorithm can be combined with the ordinary Metropolis–Hastings acceptance probability. Barker's rule becomes useful when the goal is to implement an accept–reject step through Bernoulli factories rather than through direct numerical evaluation of the acceptance probability.

2.5.2.1 The Two-Coin Algorithm

The two-coin algorithm is a basic Bernoulli factory construction used to implement Barker-type accept–reject probabilities (Gonçalves; Łatuszyński; Roberts, 2017). Suppose that the odds ratio can be factorized as

$$R(\theta, \theta') = \frac{c(\theta', \theta)p(\theta', \theta)}{c(\theta, \theta')p(\theta, \theta')}, \tag{2.41}$$

where $c(\cdot, \cdot)$ is tractable and $p(\cdot, \cdot)$ is a probability that can be simulated. Then Barker's acceptance probability can be written as

$$\alpha_B(\theta, \theta') = \frac{c(\theta', \theta)p(\theta', \theta)}{c(\theta', \theta)p(\theta', \theta) + c(\theta, \theta')p(\theta, \theta')}. \tag{2.42}$$

The two-coin algorithm generates a Bernoulli random variable with this probability without evaluating $p(\theta', \theta)$ or $p(\theta, \theta')$ numerically.

Algorithm 1: Two-coin algorithm

Flip a coin with probability proportional to $c(\theta', \theta)$ against $c(\theta, \theta')$;
if *the first-stage coin selects the proposed direction* **then**
    Flip a coin with probability $p(\theta', \theta)$;
    if *the second-stage coin is heads* **then**
        accept the proposal;
    end
else
    Flip a coin with probability $p(\theta, \theta')$;
    if *the second-stage coin is heads* **then**
        reject the proposal;
    end
if *no decision has been made* **then**
    return to the first step;
end

The algorithm loops until one of the simulated coins produces a decisive event. Conditioning on the first decisive event gives the Barker probability. Thus, the resulting accept–reject decision is exact even when the probabilities $p(\theta', \theta)$ and $p(\theta, \theta')$ are only available through simulation.

The practical difficulty is that the expected number of loops can be large. This happens when the decisive events are rare. In models with many likelihood factors, this cost can grow rapidly with the data size.

2.5.3 Divide-and-Conquer Bernoulli Factory

For models with independent observations, the likelihood often factorizes as

$$L(D \mid \theta) = \prod_{i=1}^{n} L_i(\theta). \tag{2.43}$$

A Bernoulli factory implementation may then require simulating a coin whose probability is a product of many factor probabilities. If the required probability is

$$p(\theta) = \prod_{i=1}^{n} p_i(\theta), \tag{2.44}$$

then $p(\theta)$ can become very small as n increases. A direct two-coin implementation may therefore have an expected number of loops that grows exponentially in n. This is the main computational bottleneck addressed by divide-and-conquer Bernoulli factories.

The divide-and-conquer Bernoulli factory of Stumpf-Fétizon and Gonçalves (2025) reduces this cost by recursively merging coins associated with smaller groups of factors. Instead of constructing one coin for the full product at once, the method builds a tree. The leaves correspond to individual factors or small batches of factors. The internal nodes merge the coins produced by their children. The root returns the final Bernoulli variable required by the Barker accept–reject step.

2.5.3.1 The Coin Merge Procedure

The basic merge identity can be described in terms of odds. Suppose that two independent coins have success probabilities

$$r_1 = \frac{h_1}{h_1 + g_1} \tag{2.45}$$

and

$$r_2 = \frac{h_2}{h_2 + g_2}. \tag{2.46}$$

The goal is to construct a coin with success probability

$$r_0 = \frac{h_1 h_2}{h_1 h_2 + g_1 g_2}. \tag{2.47}$$

The merge procedure repeatedly flips the two child coins until they agree. If both child coins return one, the merged coin returns one. If both child coins return zero, the merged coin returns zero. If the two child coins disagree, the procedure is repeated.

Algorithm 2: Coin merge algorithm

```
1 Flip independent coins with probabilities r_1 and r_2;
2 if both coins return the same value then
3 |   return the common value;
4 else
5 |   return to the first step;
6 end
```

The probability that the merged coin returns one is

$$\mathbb{P}(\text{return } 1) = \frac{r_1 r_2}{r_1 r_2 + (1-r_1)(1-r_2)} \tag{2.48}$$

$$= \frac{h_1 h_2}{h_1 h_2 + g_1 g_2}. \tag{2.49}$$

Thus, the merge operation multiplies the odds represented by the two child coins.

2.5.3.2 Hierarchical Factorization

Given n factors, the DCBF organizes them in a binary tree. At the leaves, each node represents either one likelihood factor or a small batch of likelihood factors. At each internal node, the two child coins are merged using Algorithm 2. The root coin represents the product of all odds contributions.

Algorithm 3: Divide-and-conquer Bernoulli factory

function *FlipDCBF*($\mathcal{F}$)
    if $\mathcal{F}$ *contains one leaf batch* **then**
        return FlipLeafCoin($\mathcal{F}$);
    end
    Split $\mathcal{F}$ into two approximately balanced subsets $\mathcal{F}_L$ and $\mathcal{F}_R$;
    $C_L \leftarrow$ FlipDCBF($\mathcal{F}_L$);
    $C_R \leftarrow$ FlipDCBF($\mathcal{F}_R$);
    return MergeCoins(C_L, C_R);
end

The role of the leaf coin is model-specific. In some applications, a leaf coin can be simulated exactly using a tractable bound. In other applications, it requires an unbiased estimator or an auxiliary construction. The exactness of the full DCBF depends on the exactness of every leaf coin and every merge operation. If a leaf coin is only approximate, then the full DCBF is also approximate.

For the normalized power prior, the DCBF can be combined with an exchange construction. The exchange step first removes the ratio $c_0(\eta)/c_0(\eta')$ by drawing $\theta^\star$ from the powered historical posterior at η'. The remaining exchange odds can be factorized over historical observations or over batches. The DCBF can then be used to simulate the Barker coin associated with this factorized exchange ratio. This route is exact only when the auxiliary exchange draw is exact and the leaf coins are exact.

2.5.3.3 Computational Complexity

The motivation for the DCBF is that a balanced tree can reduce the cost of simulating product-form Bernoulli probabilities. The vanilla two-coin algorithm may have exponential cost in the number of factors. By contrast, (Stumpf-Fétizon; Gonçalves, 2025) show that divide-and-conquer Bernoulli factory MCMC can have polynomial complexity under suitable regularity conditions. The precise polynomial degree depends on the construction of the leaf coins and on the availability of efficient unbiased estimators of the intractable likelihood ratio.

A simplified complexity statement is

$$\mathbb{E}[\mathrm{Cost}] = O\left(4^{\ell}\rho(n,\ell)\right), \tag{2.50}$$

where ℓ is the depth of the tree and $\rho(n,\ell)$ is the expected cost of the leaf-level two-coin algorithm for a batch of size $n/2^{\ell}$. When the posterior concentrates at rate $n^{-1/2}$ and the proposal is optimally scaled, suitable choices of ℓ can lead to polynomial rather than exponential complexity. This is the main theoretical advantage of the divide-and-conquer construction.

The practical message is more cautious. The DCBF is exact only if its component coins are exact. It can be computationally expensive if the merge procedure requires many repeated attempts. It also requires valid bounds or exact local constructions at the leaves. For this reason, in this thesis the DCBF is treated as a theoretically motivated exact method under bounded-support assumptions, while the exchange Metropolis–Hastings update is used as the main exact benchmark when exact powered-posterior simulation is available.

2.5.4 Pseudo-marginal Metropolis–Hastings

Pseudo-marginal Metropolis–Hastings is a modification of the Metropolis–Hastings algorithm for targets whose unnormalised density cannot be evaluated exactly, but can be estimated without bias. Let the target distribution be

$$\pi(\theta) = \frac{\gamma(\theta)}{Z}, \tag{2.51}$$

where Z is unknown and where $\gamma(\theta)$ is not available pointwise. Assume that, for each θ, we can simulate an auxiliary random variable $U \sim m_\theta(\mathrm{d}u)$ and compute a non-negative estimator $\widehat{\gamma}(\theta, U)$ satisfying

$$\mathbb{E}_{U\sim m_\theta}[\widehat{\gamma}(\theta, U)] = \gamma(\theta). \tag{2.52}$$

The pseudo-marginal method applies an ordinary Metropolis–Hastings algorithm on the extended space (θ, U). The extended target is

$$\widetilde{\pi}(\theta, u) = \frac{\widehat{\gamma}(\theta, u) m_\theta(u)}{Z}. \tag{2.53}$$

This construction is valid because the marginal distribution of θ under $\tilde{\pi}$ is the desired target distribution. Indeed,

$$\int \tilde{\pi}(\theta, u)\mathrm{d}u = \frac{1}{Z}\int \widehat{\gamma}(\theta, u) m_\theta(u)\mathrm{d}u, \tag{2.54}$$

$$= \frac{\gamma(\theta)}{Z}, \tag{2.55}$$

$$= \pi(\theta). \tag{2.56}$$

Given the current state (θ, U), the algorithm proposes $\theta' \sim q(\mathrm{d}\theta' \mid \theta)$ and independently samples $U' \sim m_{\theta'}(\mathrm{d}u')$. The proposal is accepted with probability

$$\alpha_{\mathrm{PM}}\{(\theta, U), (\theta', U')\} = \min\left\{1, \frac{\widehat{\gamma}(\theta', U')q(\theta \mid \theta')}{\widehat{\gamma}(\theta, U)q(\theta' \mid \theta)}\right\}. \tag{2.57}$$

A crucial implementation detail is that the estimator associated with the current state must be stored and reused after rejection. If the estimator at the current state were regenerated at every iteration, the resulting chain would generally no longer be the standard pseudo-marginal chain on the extended space. Under the non-negativity and unbiasedness assumptions, the algorithm is exact in the sense that the marginal invariant distribution of θ is π. This result is the central guarantee of the pseudo-marginal approach of Andrieu and Roberts (2009).

The method is especially useful when $\gamma(\theta)$ contains an intractable likelihood or an intractable normalising constant ratio for which an unbiased non-negative estimator is available. In the context of normalized power priors, one may use pseudo-marginal ideas when the ratio involving $c_0(\eta)$ is replaced by a positive unbiased estimator of the corresponding ratio.

The main caveat is that unbiasedness alone is not sufficient for good computational performance. If the estimator $\widehat{\gamma}(\theta, U)$ has high variance, the chain may become sticky because unusually large estimates at the current state are difficult to move away from. This phenomenon can produce high autocorrelation, poor effective sample size, and long burn-in periods even though the invariant distribution is theoretically correct.

Another caveat is that the estimator must be non-negative. Signed unbiased estimators do not directly define a probability density on the extended space and therefore do not yield the standard pseudo-marginal guarantee. Moreover, pseudo-marginal Metropolis–Hastings should not be confused with plugging an unbiased estimator into an arbitrary nonlinear acceptance probability. For instance, replacing an exact ratio R by an unbiased estimate $\widehat{R}$ inside Barker's probability $\widehat{R}/(1 + \widehat{R})$ is generally biased, since

$$\mathbb{E}\left[\frac{\widehat{R}}{1+\widehat{R}}\right] \neq \frac{R}{1+R}.$$

Thus, pseudo-marginal Metropolis–Hastings provides an exact route when a non-negative unbiased estimator of the unnormalised target is available, but its practical usefulness depends strongly on the stability and variance of that estimator.

2.6 Function Emulation Methods

2.6.1 Gaussian Processes

2.6.2 Shape-Constrained Additive Models

3 Approximate inference with noisy-MCMC

Current approaches:

- Function emulation: (Adame; Carvalho, 2025; Carvalho; Ibrahim, 2021)

Luiz: (Han et al., 2023) (Carvalho; Ibrahim, 2021) and references therein

Luiz: Need to carefully read, cite and discuss (Park; Haran, 2020). Do their results hold for SCAM, for instance?

Our contributions in this area:

- Do MaL estimation and talk about GP and SCAM;
- Importance sampling to simplify and improve estimation of the NPP normalising constant Adame: (including control variates on the paper);
- Showcase on moderate size examples (including NIG linear regression as a baseline). Show that it works and then show how to break it. Adame: This will go to the paper

3.1 Emulating the normalizing constant

As established in Section 2.4, the normalising constant

$$c_0(\eta) = \int_\Theta L(D_0 \mid \theta)^\eta \pi_0(\theta)\, \mathrm{d}\theta$$

also defined in (2.28) cannot be evaluated analytically in general.

As a consequence, direct evaluation of the posterior density becomes infeasible. A common strategy is therefore to replace $c_0(\eta)$ by a surrogate approximation constructed from evaluations at a finite collection of design points.

More precisely, suppose we evaluate the log-normalizing constant

$$g(\eta) = \log c_0(\eta) \tag{3.1}$$

at points $\eta_1, \ldots, \eta_m \in [0, 1]$. The evaluations may be obtained analytically, via Monte Carlo estimators, thermodynamic integration, or path sampling methods. We then construct an emulator

$$\widehat{g}(\eta) \tag{3.2}$$

intended to approximate $g(\eta)$ over the entire interval. This yields the approximation

$$c_0(\eta) \approx \exp\{\widehat{g}(\eta)\}. \tag{3.3}$$

The resulting approximation can then be plugged into the posterior density, producing an approximate MCMC scheme whose computational cost is substantially reduced after the emulator is trained.

From a statistical perspective, this transforms the problem of evaluating an intractable normalizing constant into a nonparametric regression problem. The quality of inference then depends on the approximation properties of the emulator and on how uncertainty propagates through the posterior distribution.

3.1.1 Consequences

Although emulation methods can dramatically reduce computational cost, replacing the true normalizing constant by a deterministic approximation may substantially alter the resulting posterior distribution.

Suppose the true posterior density is

$$\pi(\theta, \eta \mid D, D_0) \propto \frac{L(D \mid \theta)L(D_0 \mid \theta)^{\eta}\pi_0(\theta)\pi_0(\eta)}{c_0(\eta)}. \tag{3.4}$$

If $c_0(\eta)$ is replaced by an approximation $\widehat{c}_0(\eta)$, the resulting Markov chain targets instead

$$\widetilde{\pi}(\theta, \eta \mid D, D_0) \propto \frac{L(D \mid \theta)L(D_0 \mid \theta)^{\eta}\pi_0(\theta)\pi_0(\eta)}{\widehat{c}_0(\eta)}. \tag{3.5}$$

Consequently, approximation errors directly perturb the invariant distribution of the Markov chain. Even small systematic biases in $\widehat{c}_0(\eta)$ may produce substantial distortions in posterior summaries, credible intervals, and Bayes factors.

An important practical issue is that many emulation approaches treat $\widehat{c}_0(\eta)$ as deterministic after fitting. In this case, uncertainty associated with the emulator is ignored during posterior inference. This may lead to overconfident posterior distributions and unreliable uncertainty quantification.

The problem becomes particularly severe when the approximation error varies non-uniformly across the parameter space. For instance, local interpolation errors near regions of high posterior mass may induce systematic bias in the marginal posterior distribution of η.

Moreover, unconstrained emulators may violate structural properties known to hold for the true function. For normalized power priors, the map

$$g(\eta) = \log c_0(\eta) \tag{3.6}$$

typically satisfies smoothness and monotonicity properties inherited from the likelihood. Approximations that ignore such information may generate unrealistic artifacts, including oscillations and locally inconsistent curvature.

These issues motivate the incorporation of structural constraints into the emulation procedure.

3.1.2 Introducing shape-constraints

One strategy for improving the stability of the emulator is to incorporate qualitative information about the target function directly into the model.

Suppose

$$g(\eta) = \log c_0(\eta). \tag{3.7}$$

Under regularity conditions, derivatives of g can be expressed in terms of moments under the power posterior distribution. For instance,

$$\frac{\mathrm{d}}{\mathrm{d}\,\eta} \log c_0(\eta) = \mathbb{E}_{\pi_\eta}[\log L(D_0 \mid \theta)]. \tag{3.8}$$

Therefore, information about monotonicity and curvature may be available analytically or estimated from Monte Carlo samples.

A flexible approach consists of modelling g as a Gaussian process. If

$$g \sim \mathcal{GP}(m, k), \tag{3.9}$$

then derivatives of g are also Gaussian processes because differentiation is a linear operator. In particular,

$$g' \sim \mathcal{GP}\left(\frac{\mathrm{d}}{\mathrm{d}\,\eta} m(\eta), \frac{\mathrm{d}}{\mathrm{d}\,\eta}\frac{\mathrm{d}}{\mathrm{d}\,\eta'} k(\eta, \eta')\right). \tag{3.10}$$

This allows derivative information and shape restrictions to be incorporated directly into the posterior distribution of the Gaussian process. Conditioning on derivative observations or derivative sign constraints can substantially reduce unrealistic oscillatory behaviour. We refer to Riihimaki and Vehtari (2010), Wang and Berger (2016) and Adame and Carvalho (2025) for recent developments in shape-constrained emulation through Gaussian processes.

An alternative strategy is based on shape-constrained additive models (SCAMs). These models extend generalized additive models (GAMs) by imposing constraints directly on the spline basis coefficients.

Suppose

$$g(\eta) = \beta_0 + \sum_{j=1}^{p} f_j(\eta), \tag{3.11}$$

where the functions f_j are represented through spline expansions. Shape constraints such as monotonicity or convexity can then be enforced by restricting the spline coefficients during estimation.

In practice, SCAMs are fitted using penalized likelihood methods analogous to standard GAM estimation procedures. The main difference is that the optimization is performed under linear inequality constraints on the spline coefficients.

Compared with unconstrained smoothers, shape-constrained models may provide more stable extrapolation behaviour and improved uncertainty quantification in regions with sparse design points. These properties are particularly appealing when approximating normalizing constants, since small local errors may propagate nonlinearly into the posterior distribution.

3.2 Current approaches for the NPP

The normalised power prior has received considerable attention in recent years due to its ability to incorporate historical information while preserving coherence with the likelihood principle. However, as established in Section 2.4, the normalising constant $c_0(\eta)$ depends on η in a non-analytic manner for most models of interest, inducing a doubly intractable posterior whose evaluation requires repeated computation of $c_0(\eta)$ across MCMC iterations.

A substantial portion of the current literature therefore focuses on approximating either the normalizing constant itself or ratios involving $c_0(\eta)$. The main approaches may be broadly divided into:

(i) direct approximation of the normalizing function;

(ii) Monte Carlo estimators based on importance sampling or path sampling; and

(iii) exact or asymptotically exact MCMC procedures.

3.2.1 Approximate inference through emulation and interpolation

One of the main practical difficulties in the NPP is that evaluating $c_0(\eta)$ repeatedly during posterior simulation may become prohibitively expensive for complex models. Consequently, several authors have proposed constructing surrogate approximations for the normalizing function.

A common strategy consists of evaluating $c_0(\eta)$ or $\log c_0(\eta)$ over a finite grid of discounting parameters and then interpolating these values over the interval $\eta \in [0,1]$. Spline-based approximations are particularly attractive in this setting because the normalizing function is typically smooth as a function of η.

In the R package NPP associated with Han et al. (2023), efficient posterior computation is achieved by avoiding repeated evaluation of the normalizing constant during MCMC sampling. The proposed framework exploits structural properties of the power posterior and constructs computational approximations that only require simulation from a small number of reference distributions.

More generally, spline interpolation methods approximate

$$g(\eta) = \log c_0(\eta) \tag{3.12}$$

through basis expansions of the form

$$\widehat{g}(\eta) = \sum_{k=1}^{K} \beta_k B_k(\eta), \tag{3.13}$$

where B_k denotes spline basis functions. The resulting approximation can then be plugged directly into the posterior distribution.

Although such methods may substantially reduce computational cost, they typically treat the approximation as deterministic after fitting. As a consequence, uncertainty associated with the approximation is not propagated into the posterior distribution. This may lead to overconfident inference and distortions in posterior summaries when approximation errors accumulate near regions of high posterior mass.

An important limitation of the current literature is that there is still no widely adopted exact computational baseline for assessing approximation quality in the NPP setting. Most existing works compare computational approximations against other approximate procedures or against simplified models where the normalizing constant is analytically available. This makes it difficult to quantify the true inferential impact of approximation errors.

The recent development of exact and quasi-exact MCMC procedures for doubly intractable distributions provides an opportunity to revisit these approximation methods under a more rigorous computational framework. In particular, exact samplers may be used as gold-standard references for evaluating the bias and stability of surrogate models.

3.2.2 Monte Carlo approaches and exact inference

Several recent works investigate Monte Carlo estimators for posterior inference under the normalized power prior. These methods typically exploit the representation of the normalizing constant as an expectation under a tractable proposal distribution.

The work of Carvalho and Ibrahim (2021) establishes several theoretical properties of the NPP and highlights the computational challenges induced by the normalizing function. In particular, the authors show that the normalized power prior is always proper whenever the initial prior is proper and prove smoothness and convexity properties of the map $\eta \mapsto \log c_0(\eta)$.

The same work also emphasizes that posterior inference under the NPP belongs to the class of doubly intractable distributions because the posterior density depends explicitly on an unknown normalizing constant ratio. This observation motivates the use of advanced MCMC procedures originally developed for doubly intractable models.

More recently, Han et al. (2023) proposed efficient posterior sampling procedures based on importance sampling and power posterior representations. The central idea is to avoid evaluating $c_0(\eta)$ repeatedly by constructing estimators using samples generated under reference power posterior distributions.

These approaches are computationally attractive because they are relatively simple to implement and scale well to moderate-dimensional models. However, they remain approximate in general and their accuracy depends critically on the variance of the Monte Carlo estimators and on the overlap between the proposal and target distributions.

At present, the literature still lacks a systematic comparison between:

(i) deterministic emulation procedures;

(ii) pseudo-marginal Monte Carlo estimators; and

(iii) exact MCMC algorithms for the NPP.

In particular, there is limited empirical evidence regarding: the stationary bias induced by noisy approximations, the computational trade-offs between exact and approximate methods, and the practical benefits of incorporating structural information such as shape constraints into surrogate models.

The methodology developed in this work aims to contribute to this direction by combining exact MCMC procedures based on the Exchange algorithm with approximation strategies based on functional emulation. This provides a unified framework in which both exact and approximate methods can be evaluated relative to a common computational baseline.

3.3 Doubly Intractable Settings

Adame: Discuss (Park; Haran, 2020)

As established in Section 2.3, doubly intractable distributions arise when the posterior density contains a normalising constant that depends on the parameter being updated and cannot be evaluated pointwise. The normalised power prior is a canonical example of this phenomenon. This section reviews the main computational strategies available in this setting, with a focus on the emulation approach of Park and Haran (2020) and its applicability to the NPP.

A number of Markov chain Monte Carlo (MCMC) methods have been proposed to address this problem, including auxiliary variable approaches such as the exchange algorithm and the double Metropolis–Hastings (DMH) algorithm. While these methods can target the exact posterior distribution, they typically require simulating auxiliary variables at each iteration, which becomes computationally prohibitive in high-dimensional or complex models.

An alternative strategy is proposed by Park and Haran (2020), who introduce a function emulation approach to approximate the intractable normalizing constant. Their method replaces repeated Monte Carlo estimation of $Z(\theta)$ with a two-stage surrogate model. First, importance sampling is used to obtain noisy estimates of $Z(\theta)$ at a finite set of parameter values. Then, a Gaussian process (GP) is fitted to the log-normalizing function, providing a fast interpolator that can be evaluated within an MCMC algorithm. This leads to two variants: one that emulates only the normalizing function (NormEm), and another that emulates the entire likelihood (LikEm).

The main advantage of this approach is computational efficiency. By shifting the costly Monte Carlo approximations to a precomputation stage, the resulting MCMC algorithm avoids repeated simulation of auxiliary variables, yielding substantial speed-ups compared to exact or pseudo-marginal methods. Moreover, the approach is naturally parallelizable and extends beyond doubly intractable settings to more general problems involving expensive likelihood evaluations.

However, this efficiency comes at the cost of asymptotic exactness. The resulting Markov chain targets an approximate posterior distribution, with an error that depends on both the number of importance samples and the number of design points used to train the Gaussian process. While theoretical bounds on the total variation distance are provided, they do not yield explicit rates, making it difficult to calibrate the approximation in practice.

Several additional limitations remain. First, the method relies on importance sampling estimates, which can exhibit high variance in moderate to high-dimensional settings, potentially degrading the quality of the surrogate. Second, the Gaussian process approximation assumes a smooth structure for the log-normalizing function and typically ignores heteroskedasticity induced by Monte Carlo noise. Third, the selection of design points (particles) is critical but performed using heuristic procedures such as short runs of DMH or approximate Bayesian computation, without a principled adaptive strategy. Finally, the approach does not scale well with the dimension of the parameter space due to the cubic complexity of Gaussian process regression and the increasing difficulty of covering the parameter space.

These limitations suggest several directions for future research. In particular, combining surrogate-based approximations with exact correction mechanisms, such as pseudo-marginal or delayed-acceptance schemes, could potentially recover asymptotic exactness while retaining computational gains. Additionally, replacing importance sampling with more robust estimators, and incorporating adaptive or local surrogate models, may improve both stability and scalability. Such developments are particularly relevant in the context of NPP models, where efficient and reliable inference remains an open challenge.

4 Exact inference for the Normalised Power Prior

In this chapter, we develop exact Markov chain Monte Carlo methods for inference under the normalised power prior.

The full joint posterior of interest is

$$\pi(\theta, \eta \mid D, D_0) \propto L(D \mid \theta)\, \frac{L_0(D_0 \mid \theta)^{\eta}\, \pi_0(\theta)}{c_0(\eta)}\, \pi_A(\eta), \tag{4.1}$$

and the central challenge is the intractable ratio $c_0(\eta)/c_0(\eta')$ that appears in any Metropolis–Hastings update of η.

We address this challenge through three complementary constructions: the exchange algorithm (Section 4.1), the Poisson estimator (Section 4.2.1), and a pseudo-marginal approach based on importance sampling (Section 4.3).

Updating θ given (η, D, D_0) is straightforward: the full conditional

$$\pi(\theta \mid \eta, D, D_0) \propto L(D \mid \theta)\, L_0(D_0 \mid \theta)^{\eta}\, \pi_0(\theta) \tag{4.2}$$

does not involve $c_0(\eta)$, so standard Metropolis–Hastings (Section 2.2.2) applies directly.

For a symmetric proposal $q(\theta' \mid \theta) = q(\theta \mid \theta')$, the acceptance ratio reduces to

$$\alpha(\theta, \theta' \mid \eta) = \min\left\{1,\ \frac{L(D \mid \theta')\, \pi_0(\theta')}{L(D \mid \theta)\, \pi_0(\theta)} \left(\frac{L_0(D_0 \mid \theta')}{L_0(D_0 \mid \theta)}\right)^{\eta}\right\}. \tag{4.3}$$

The remainder of this chapter focuses on the harder problem of updating η.

4.1 Exchange Metropolis–Hastings

In order to apply the Exchange Algorithm, as presented in section 2.5.1, we find the intractable quantity to be

$$c_0(\eta) = \int_{\Theta} L_0(D_0 \mid \theta)^{\eta} \pi_0(\theta) \mathrm{d}\theta. \tag{4.4}$$

If the current value of θ is fixed and a proposal η' is generated from $q(\eta' \mid \eta)$, the exchange update draws

$$\theta^{\star} \sim \pi_{\eta'}(\theta \mid D_0) = \frac{L_0(D_0 \mid \theta)^{\eta'} \pi_0(\theta)}{c_0(\eta')}. \tag{4.5}$$

The exchange ratio for the η update is then

$$R_{\mathrm{ex}}(\eta, \eta'; \theta, \theta^\star) = \frac{\pi_A(\eta')}{\pi_A(\eta)} \frac{q(\eta \mid \eta')}{q(\eta' \mid \eta)} \frac{L_0(D_0 \mid \theta)^{\eta'-\eta}}{L_0(D_0 \mid \theta^\star)^{\eta'-\eta}}. \tag{4.6}$$

The ratio $c_0(\eta)/c_0(\eta')$ has disappeared. This cancellation is the main reason why the exchange algorithm is attractive for the normalized power prior.

`Adame:` Add proof and intuition that it is not biased. According to the exchange algorithm framework of Murray, Ghahramani and MacKay (2012), the auxiliary draw from $\pi_{\eta'}(\theta \mid D_0)$ introduces a factor $1/c_0(\eta')$ into the augmented target that exactly cancels the intractable ratio (see Appendix A.3 for a self-contained argument).

The exactness of the exchange algorithm depends on the ability to sample $\theta^\star$ exactly from $\pi_{\eta'}(\theta \mid D_0)$. This assumption is strong. It is satisfied in conjugate Gaussian models and in some conditionally conjugate models. It is not automatically satisfied in general regression models or in non-conjugate likelihoods. If $\theta^\star$ is generated by an inner MCMC chain that is not run to stationarity, the resulting algorithm is no longer exactly the exchange algorithm. In that case, the method becomes an approximate exchange or double Metropolis–Hastings algorithm.

4.2 Barker's Method via Poisson Estimator

Updating η is more delicate because the full conditional

$$\pi(\eta \mid \theta, D, D_0) \propto \frac{\pi_A(\eta)}{c_0(\eta)} L_0(D_0 \mid \theta)^\eta \tag{4.7}$$

involves $c_0(\eta)$ explicitly.

The Metropolis–Hastings ratio for a proposal $\eta \to \eta'$ is

$$\frac{\pi(\eta' \mid \theta)}{\pi(\eta \mid \theta)} = \frac{\pi_A(\eta')}{\pi_A(\eta)} \cdot \frac{c_0(\eta)}{c_0(\eta')} \cdot L_0(D_0 \mid \theta)^{\eta'-\eta}, \tag{4.8}$$

and the intractable ratio $c_0(\eta)/c_0(\eta')$ prevents direct evaluation.

Recalling the thermodynamic identity established in (2.30), integrating from η to η' gives

$$\Delta(\eta, \eta') := \log \frac{c_0(\eta')}{c_0(\eta)} = \int_\eta^{\eta'} \mathbb{E}_{\pi_\tau}\left[\log L_0(D_0 \mid \theta)\right] \mathrm{d}\tau, \tag{4.9}$$

where $\pi_\tau(\theta) \propto L_0(D_0 \mid \theta)^\tau \pi_0(\theta)$ is the power prior at temperature τ.

The goal of the following sections is to construct a non-negative unbiased estimator of $\exp\{-\Delta(\eta, \eta')\} = c_0(\eta)/c_0(\eta')$.

4.2.1 Producing an unbiased estimator via a Poisson estimator

Let π_τ denote the power posterior associated with the historical data. For $\tau \in [0,1]$, define

$$\pi_\tau(\theta) = \frac{L_0(D_0 \mid \theta)^\tau \pi_0(\theta)}{c_0(\tau)}, \qquad c_0(\tau) = \int_\Theta L_0(D_0 \mid t)^\tau \, \mathrm{d}P_0(t). \tag{4.10}$$

For two values $\eta, \eta' \in [0,1]$, define

$$\Delta(\eta, \eta') = \log \frac{c_0(\eta')}{c_0(\eta)}. \tag{4.11}$$

Under the usual regularity conditions allowing differentiation under the integral sign, the thermodynamic identity gives

$$\Delta(\eta, \eta') = \int_\eta^{\eta'} \mathbb{E}_{\theta \sim \pi_\tau} \left[\log L_0(D_0 \mid \theta)\right] \, \mathrm{d}\tau. \tag{4.12}$$

Equivalently, if $U \sim \mathrm{Unif}(0,1)$ and $\tau_{\eta,\eta'}(U) = \eta + U(\eta' - \eta)$, then

$$\Delta(\eta, \eta') = \mathbb{E}_U \left[(\eta' - \eta) \mathbb{E}_{\theta \sim \pi_{\tau_{\eta,\eta'}(U)}} \{\log L_0(D_0 \mid \theta)\} \right]. \tag{4.13}$$

This representation suggests a proposal-specific random variable. Let $\mathsf{P}_{\eta,\eta'}$ be the probability measure on $[0,1] \times \Theta$ obtained by first drawing $U \sim \mathrm{Unif}(0,1)$ and then drawing $\theta \sim \pi_{\tau_{\eta,\eta'}(U)}$. Equivalently,

$$\mathsf{P}_{\eta,\eta'}(\mathrm{d}u, \mathrm{d}\theta) = \mathrm{d}u \, \pi_{\tau_{\eta,\eta'}(u)}(\mathrm{d}\theta). \tag{4.14}$$

Define

$$Z_{\eta,\eta'}(U, \theta) = (\eta' - \eta) \log L_0(D_0 \mid \theta). \tag{4.15}$$

Then

$$\mathbb{E}_{\mathsf{P}_{\eta,\eta'}}[Z_{\eta,\eta'}] = \Delta(\eta, \eta'). \tag{4.16}$$

The objective is to construct an unbiased estimator of the ratio $c_0(\eta)/c_0(\eta')$. This ratio can be written as

$$\exp\{-\Delta(\eta, \eta')\} = \frac{c_0(\eta)}{c_0(\eta')}. \tag{4.17}$$

Assume that there exist deterministic constants $a_{\eta,\eta'}$ and $b_{\eta,\eta'}$ such that

$$a_{\eta,\eta'} \leq Z_{\eta,\eta'}(U, \theta) \leq b_{\eta,\eta'} \qquad \mathsf{P}_{\eta,\eta'}\text{-almost surely}. \tag{4.18}$$

Set

$$c_{\eta,\eta'} = b_{\eta,\eta'}, \qquad \lambda_{\eta,\eta'} = b_{\eta,\eta'} - a_{\eta,\eta'}. \tag{4.19}$$

Draw $K \sim \mathrm{Poisson}(\lambda_{\eta,\eta'})$. Conditionally on K, draw

$$(U_j, \theta_j)_{j=1}^K \overset{\text{iid}}{\sim} \mathsf{P}_{\eta,\eta'}. \tag{4.20}$$

The Poisson estimator is

$$\widehat{I}(\eta, \eta') = \exp\{\lambda_{\eta,\eta'} - c_{\eta,\eta'}\} \prod_{j=1}^{K} \frac{c_{\eta,\eta'} - Z_{\eta,\eta'}(U_j, \theta_j)}{\lambda_{\eta,\eta'}}. \tag{4.21}$$

By construction, each factor satisfies

$$0 \leq \frac{c_{\eta,\eta'} - Z_{\eta,\eta'}(U_j, \theta_j)}{\lambda_{\eta,\eta'}} \leq 1. \tag{4.22}$$

Therefore $\widehat{I}(\eta, \eta')$ is non-negative. Using the exponential formula for a Poisson random variable, we obtain

$$\begin{aligned}
\mathbb{E}[\widehat{I}(\eta, \eta')] &= \exp\{\lambda_{\eta,\eta'} - c_{\eta,\eta'}\}\mathbb{E}\left[\left\{\mathbb{E}_{\mathsf{P}_{\eta,\eta'}}\left[\frac{c_{\eta,\eta'} - Z_{\eta,\eta'}}{\lambda_{\eta,\eta'}}\right]\right\}^K\right] &(4.23)\\
&= \exp\{\lambda_{\eta,\eta'} - c_{\eta,\eta'}\} \exp\left\{\lambda_{\eta,\eta'}\left(\frac{c_{\eta,\eta'} - \Delta(\eta, \eta')}{\lambda_{\eta,\eta'}} - 1\right)\right\} &(4.24)\\
&= \exp\{-\Delta(\eta, \eta')\}. &(4.25)
\end{aligned}$$

Thus, $\widehat{I}(\eta, \eta')$ is an unbiased estimator of the normalizing-constant ratio $c_0(\eta)/c_0(\eta')$.

The efficiency of the estimator depends strongly on the envelope width $\lambda_{\eta,\eta'} = b_{\eta,\eta'} - a_{\eta,\eta'}$. Large envelopes lead to large Poisson counts and high estimator variability. Therefore, the practical usefulness of the construction depends on obtaining deterministic but sufficiently tight proposal-specific bounds for $Z_{\eta,\eta'}$.

It is important to distinguish this unbiased estimator from an exact Barker accept–reject mechanism. Although $\widehat{I}(\eta, \eta')$ is unbiased for $c_0(\eta)/c_0(\eta')$, substituting a noisy estimate of the likelihood ratio directly into the Barker probability does not generally produce an exact Barker transition. Exact Barker implementations require simulating the accept–reject event itself with the correct marginal probability, typically through a Bernoulli-factory or two-coin construction.

4.2.1.1 Practical implementation

The integral $\Delta(\eta, \eta')$ in (4.9) is estimated by Monte Carlo.

Drawing $\tau_k \overset{\text{iid}}{\sim} \text{Unif}(\eta, \eta')$ and, conditionally, $\widetilde{\theta}_{m,k} \sim \pi_{\tau_k}$ for $m = 1, \ldots, M$ and $k = 1, \ldots, K$, gives the double Monte Carlo estimator

$$\widehat{\Delta}(\eta, \eta') = \frac{(\eta' - \eta)}{MK} \sum_{k=1}^{K} \sum_{m=1}^{M} \log L_0\Big(D_0 \mid \widetilde{\theta}_{m,k}\Big). \tag{4.26}$$

This is an unbiased estimator of $\Delta(\eta, \eta')$ by the law of total expectation.

In the single-draw case ($K = M = 1$), this reduces to

$$\widehat{\Delta}(\eta, \eta') = (\eta' - \eta) \log L_0(D_0 \mid \widetilde{\theta}_\star), \qquad \tau_\star \sim \text{Unif}(\eta, \eta'), \quad \widetilde{\theta}_\star \sim \pi_{\tau_\star}, \tag{4.27}$$

which can be used in the Poisson estimator construction of Section 4.2.1.

Note that only the mean of $Z_{\eta,\eta'}$, not its variance, enters the unbiasedness argument; the variance governs the efficiency of the estimator through the expected Poisson count $\lambda_{\eta,\eta'}$.

4.2.2 Local envelopes for the Poisson estimator

The preceding construction does not require a single global envelope that is valid uniformly over all possible proposals. Instead, for each proposed move $\eta \to \eta'$, it is enough to obtain deterministic bounds for the proposal-specific random variable

$$Z_{\eta,\eta'}(U,\theta) = (\eta' - \eta) \log L_0(D_0 \mid \theta), \qquad (U,\theta) \sim \mathsf{P}_{\eta,\eta'}. \tag{4.28}$$

Suppose that the historical log-likelihood satisfies

$$\ell_{\min} \le \log L_0(D_0 \mid \theta) \le \ell_{\max} \qquad \text{for all } \theta \in \Theta. \tag{4.29}$$

Then, for a fixed proposal $\eta \to \eta'$, define

$$a_{\eta,\eta'} = \min\{(\eta' - \eta)\ell_{\min}, (\eta' - \eta)\ell_{\max}\}. \tag{4.30}$$

Similarly, define

$$b_{\eta,\eta'} = \max\{(\eta' - \eta)\ell_{\min}, (\eta' - \eta)\ell_{\max}\}. \tag{4.31}$$

The resulting envelope width is

$$b_{\eta,\eta'} - a_{\eta,\eta'} = |\eta' - \eta|(\ell_{\max} - \ell_{\min}). \tag{4.32}$$

Hence a natural local choice is

$$c_{\eta,\eta'} = b_{\eta,\eta'}, \qquad \lambda_{\eta,\eta'} = |\eta' - \eta|(\ell_{\max} - \ell_{\min}). \tag{4.33}$$

This choice makes the expected number of Poisson factors proportional to the proposal distance $|\eta' - \eta|$. Consequently, local random-walk proposals in η lead to smaller expected computational cost than large independent proposals.

For unbounded parameter spaces, such as Gaussian models with $\Theta = \mathbb{R}^d$, the lower bound $\ell_{\min}$ is typically $-\infty$. In that case, the above exact Poisson construction is not directly applicable without additional structure. One possible solution is to work with a compactly supported prior. For example, define

$$\pi_{0,B}(\theta) \propto \pi_0(\theta)\mathbf{1}_{\Theta_B}(\theta), \tag{4.34}$$

where $\Theta_B \subset \Theta$ is compact. Then deterministic bounds for $\log L_0(D_0 \mid \theta)$ can be derived on Θ_B. In the Gaussian historical likelihood with known covariance, the log-likelihood is a concave quadratic function of θ. Its maximum is attained at the historical maximum likelihood estimator. Its minimum over a compact box is attained on the boundary. These deterministic bounds yield a valid local envelope for the Poisson estimator.

4.2.3 Divide-and-conquer factorization

Adame: Move it to an appendix? I'd say at least the proof should be in proofs.

The computational difficulty of the Poisson estimator increases with the variability of the historical log-likelihood. Since the historical likelihood factorizes over observations, we can write

$$\log L_0(D_0 \mid \theta) = \sum_{i=1}^{N_0} \log L_i(d_{0i} \mid \theta). \tag{4.35}$$

The width of a global envelope for $Z_{\eta,\eta'} = (\eta' - \eta) \log L_0(D_0 \mid \theta)$ typically grows with N_0. This motivates a divide-and-conquer construction.

Let $D_{0,1}, \dots, D_{0,G}$ be a partition of the historical data. Let $\mathcal{I}_g$ denote the set of observation indices in block g. Define the block likelihood by

$$L_{0,g}(D_{0,g} \mid \theta) = \prod_{i \in \mathcal{I}_g} L_i(d_{0i} \mid \theta). \tag{4.36}$$

Then the full historical likelihood satisfies

$$L_0(D_0 \mid \theta) = \prod_{g=1}^{G} L_{0,g}(D_{0,g} \mid \theta). \tag{4.37}$$

Equivalently, the full historical log-likelihood satisfies

$$\log L_0(D_0 \mid \theta) = \sum_{g=1}^{G} \log L_{0,g}(D_{0,g} \mid \theta). \tag{4.38}$$

For each block g, define the blockwise random variable

$$Z_{g,\eta,\eta'}(U, \theta) = (\eta' - \eta) \log L_{0,g}(D_{0,g} \mid \theta). \tag{4.39}$$

Assume that deterministic bounds $a_{g,\eta,\eta'}$ and $b_{g,\eta,\eta'}$ are available such that

$$a_{g,\eta,\eta'} \le Z_{g,\eta,\eta'}(U, \theta) \le b_{g,\eta,\eta'} \qquad \mathsf{P}_{\eta,\eta'}\text{-almost surely.} \tag{4.40}$$

Set

$$c_{g,\eta,\eta'} = b_{g,\eta,\eta'}, \qquad \lambda_{g,\eta,\eta'} = b_{g,\eta,\eta'} - a_{g,\eta,\eta'}. \tag{4.41}$$

A blockwise Poisson estimator can then be constructed as

$$\widehat{I}_g(\eta, \eta') = \exp\{\lambda_{g,\eta,\eta'} - c_{g,\eta,\eta'}\} \prod_{j=1}^{K_g} \frac{c_{g,\eta,\eta'} - Z_{g,\eta,\eta'}(U_{g,j}, \theta_{g,j})}{\lambda_{g,\eta,\eta'}}, \tag{4.42}$$

where $K_g \sim \text{Poisson}(\lambda_{g,\eta,\eta'})$. The variables $(U_{g,j}, \theta_{g,j})$ are sampled independently from the same proposal-specific measure $\mathsf{P}_{\eta,\eta'}$. The blockwise estimator satisfies

$$\mathbb{E}[\widehat{I}_g(\eta, \eta')] = \exp\{-\Delta_g(\eta, \eta')\}, \tag{4.43}$$

where

$$\Delta_g(\eta, \eta') = \mathbb{E}_{\mathsf{P}_{\eta,\eta'}} \left[Z_{g,\eta,\eta'}(U, \theta) \right]. \tag{4.44}$$

If the block estimators are independent, the divide-and-conquer product estimator is

$$\widehat{I}_{\mathrm{dc}}(\eta, \eta') = \prod_{g=1}^{G} \widehat{I}_g(\eta, \eta'). \tag{4.45}$$

By independence, its expectation is

$$\mathbb{E}[\widehat{I}_{\mathrm{dc}}(\eta, \eta')] = \prod_{g=1}^{G} \mathbb{E}[\widehat{I}_g(\eta, \eta')] \tag{4.46}$$

$$= \prod_{g=1}^{G} \exp\{-\Delta_g(\eta, \eta')\} \tag{4.47}$$

$$= \exp\left\{ -\sum_{g=1}^{G} \Delta_g(\eta, \eta') \right\}. \tag{4.48}$$

Since

$$\sum_{g=1}^{G} \Delta_g(\eta, \eta') = \Delta(\eta, \eta'), \tag{4.49}$$

we obtain

$$\mathbb{E}[\widehat{I}_{\mathrm{dc}}(\eta, \eta')] = \exp\{-\Delta(\eta, \eta')\}. \tag{4.50}$$

Thus, the divide-and-conquer product estimator is unbiased for $c_0(\eta)/c_0(\eta')$.

The main advantage of this construction is variance and cost control. Instead of using one global envelope with width $\lambda_{\eta,\eta'}$, the method uses several smaller envelopes with widths $\lambda_{g,\eta,\eta'}$. The expected number of Poisson factors is

$$\sum_{g=1}^{G} \lambda_{g,\eta,\eta'}. \tag{4.51}$$

When the blockwise envelopes are much tighter than the full-data envelope, the product construction can be substantially more stable than the global estimator.

This product estimator should not be confused with a genuine recursive divide-and-conquer Bernoulli factory. The estimator $\widehat{I}_{\mathrm{dc}}(\eta, \eta')$ is an unbiased estimator of a normalizing-constant ratio. However, using it as a noisy plug-in inside the Barker probability does not generally yield an exact Barker transition. A true DCBF Barker transition recursively constructs the accept–reject coin itself. At the leaves, one simulates smaller Bernoulli coins associated with subsets of likelihood factors. At internal nodes, these coins are merged recursively until a single coin with the Barker acceptance probability is obtained at the root. Therefore, the recursive DCBF is a coin construction, not merely a product of random ratio estimates.

4.3 Pseudo-marginal via Importance Sampling

An alternative to constructing an unbiased estimator of the ratio $c_0(\eta)/c_0(\eta')$ is to directly estimate the normalizing constant $c_0(\eta)$ itself and form the ratio from two independent estimates.

The simplest such approach is the *prior Monte Carlo estimator*, which exploits the representation

$$c_0(\eta) = \mathbb{E}_{\theta\sim\pi_0}[L_0(\theta \mid D_0)^\eta].$$

Samples drawn directly from the prior π_0 are cheap to obtain and require no gradient information or auxiliary computations. The main limitation is that the prior and the power posterior $\pi_\eta(\theta) \propto L_0(\theta \mid D_0)^\eta \pi_0(\theta)$ can be poorly aligned, particularly when the historical data are informative, leading to high variance in the importance weights and an inefficient estimator.

A more principled alternative is *importance sampling* centered at the maximum likelihood estimate. Writing the normalizing constant as

$$c_0(\eta) = \int_{\theta\in\Theta} L_0(\theta \mid D_0)^\eta \pi_0(\theta)\,\mathrm{d}\theta = \mathbb{E}_{\theta\sim\pi_{\mathrm{MLE}}}\left[L_0(\theta \mid D_0)^\eta \frac{\pi_0(\theta)}{\pi_{\mathrm{MLE}}(\theta)}\right],$$

one draws samples from a proposal π_{MLE} centered at the MLE and reweights by the ratio $\pi_0(\theta)/\pi_{\mathrm{MLE}}(\theta)$. The MLE-centered proposal concentrates mass where the likelihood is large, reducing variance relative to the prior estimator when the historical data are informative. The cost is the need to compute or approximate the MLE, which may itself require numerical optimization.

5 Experiments

5.1 Assessing exact and approximate samplers

The primary objective of this chapter is to compare exact and approximate inference procedures for the normalized power prior under controlled settings where the normalizing constant is analytically available.

This allows us to evaluate not only computational efficiency, but also the inferential distortion introduced by noisy approximations of the normalizing constant ratio.

A central distinction throughout the experiments is between:

(i) exact MCMC algorithms targeting the true posterior distribution; and

(ii) pseudo-marginal or approximation-based methods whose stationary distribution depends on Monte Carlo estimators or surrogate approximations.

Rather than comparing methods exclusively through asymptotic guarantees, we focus on their finite-sample practical behaviour.

In particular, we evaluate posterior agreement with the analytic baseline, computational efficiency, acceptance rates, Monte Carlo variability, and stability of the normalising-constant estimators.

Since the posterior distribution of η is one-dimensional, we use the empirical Wasserstein distance relative to the exact analytic sampler as the primary discrepancy measure.

Unlike density-based divergences, the Wasserstein distance does not require kernel density estimation and remains numerically stable even when posterior samples concentrate near the boundary of the parameter space.

5.1.1 Using Exact Methods as Baseline

The Exchange MH algorithm described in Section 4.1 serves as the primary exact reference throughout the experiments. Where the powered prior $\pi_{\eta'}(\theta \mid D_0)$ is available in closed form (Gaussian and Normal–Inverse–Gamma models), the exchange update is exact and computationally efficient.

The importance sampling estimators δ_a (prior IS) and δ_b (MLE-centred IS), introduced in Section 4.3, are used as pseudo-marginal baselines. They provide non-negative unbiased estimates of $c_0(\eta)$ and can be embedded within a Metropolis–Hastings scheme, yielding exact pseudo-marginal chains under the conditions of Andrieu and Roberts (2009).

5.2 Convergence to the baseline

5.2.1 Normal distribution with known covariance

We evaluate the proposed samplers under a multivariate Gaussian model with known covariance structure. The parameter dimension is fixed at $p = 3$, with

$$\Sigma = \boldsymbol{I}_3, \qquad \Sigma_0 = 2\boldsymbol{I}_3. \tag{5.1}$$

A Gaussian prior is assigned to θ,

$$\theta \sim \mathcal{N}_3(\mathbf{0}, \Sigma_0), \tag{5.2}$$

and the power parameter receives a uniform prior,

$$\eta \sim \text{Beta}(1, 1). \tag{5.3}$$

Historical and current datasets each contain $N_0 = N = 50$ observations.

Three discrepancy scenarios are considered. In the small-discrepancy regime, the historical and current datasets are generated from nearby means. The medium-discrepancy regime introduces a moderate shift between the datasets, while the large-discrepancy regime generates substantial disagreement between the historical and current information. As the discrepancy increases, the posterior distribution of η is expected to concentrate closer to zero, reflecting reduced borrowing from the historical data.

The purpose of this experiment is not only to compare posterior distributions, but also to assess the computational efficiency required for each method to approximate the exact posterior distribution. To accomplish this, we use a convergence-to-baseline protocol.

The analytic Metropolis–Hastings sampler is used as the reference method. Since the Gaussian normalized power prior admits a closed-form normalizing constant ratio, this sampler directly evaluates the exact posterior density. Longer chains are used for the analytic reference in order to reduce Monte Carlo variability and provide a stable empirical baseline for posterior comparisons.

The remaining methods are evaluated according to how efficiently they approach the analytic reference posterior. We compare four methods: Exchange MH, Prior IS, MLE-centred IS, and Path Sampling PMMH.

For each method, chains are accumulated until the one-dimensional Wasserstein distance between the empirical posterior distribution of η and the analytic reference posterior falls below a fixed tolerance threshold. We additionally monitor the Wasserstein distance for $\|\theta\|_2$ in order to assess agreement in the parameter space.

The primary performance metric is therefore the computational effort required to achieve a prescribed approximation accuracy. We report the Wasserstein distance to the analytic baseline, total runtime, effective sample size, posterior means and standard deviations, and acceptance rates.

The Wasserstein distance is preferred over the Kolmogorov–Smirnov statistic because it provides a more stable and geometrically meaningful measure of posterior discrepancy, particularly for concentrated or skewed distributions.

A central distinction between the methods is how the ratio of normalizing constants is handled. The Exchange MH algorithm avoids evaluating the normalizing constant entirely by introducing an auxiliary draw from the powered prior distribution at the proposed value of η. In the present Gaussian experiment, sampling from the powered prior can be performed exactly in closed form, making the exchange update computationally efficient while preserving exactness.

The Prior IS and MLE-centered IS methods instead estimate the normalizing constant ratio through importance sampling constructions. Both methods remain exact pseudo-marginal algorithms when embedded within the extended-state Metropolis–Hastings framework, provided the estimator is nonnegative and unbiased. The difference between the two approaches lies primarily in variance reduction. The MLE-centered proposal attempts to concentrate samples in regions of high likelihood in order to stabilize the importance weights.

The Path Sampling PMMH method approximates the log normalizing constant ratio through thermodynamic integration along a temperature path. Although effective in some settings, the resulting estimator is not guaranteed to satisfy the conditions required for exact pseudo-marginal inference and is therefore treated as an approximate baseline.

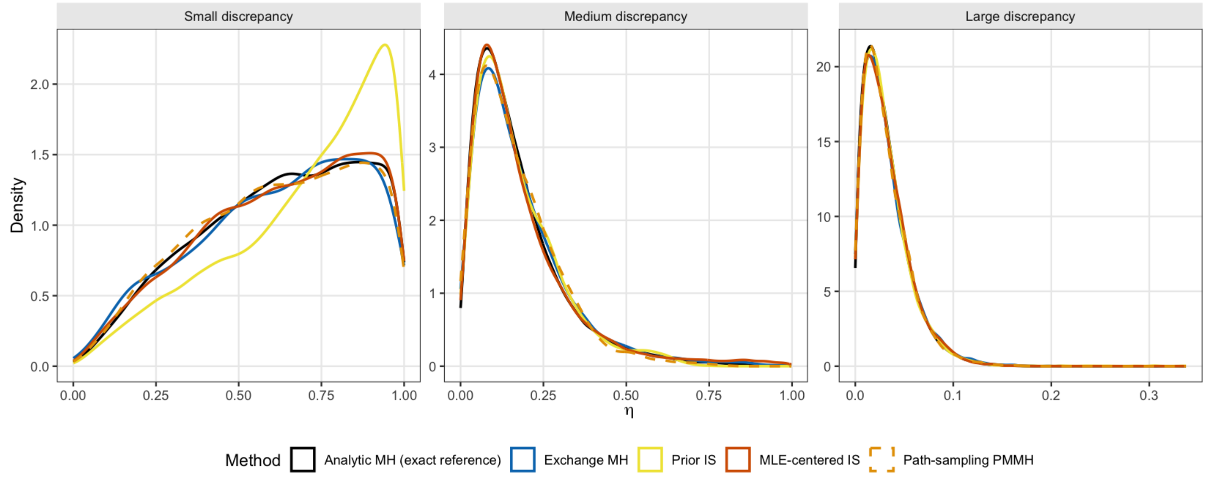

Figure 1 – Posterior density of η across discrepancy regimes for the Gaussian normalized power prior experiment.

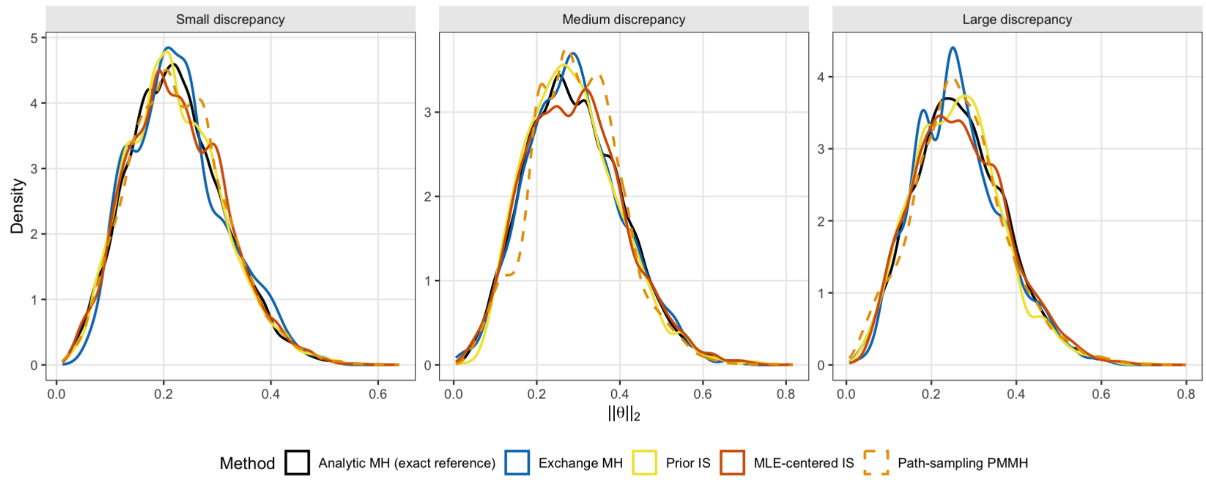

Figure 2 – Posterior density of $\|\theta\|_2$ across discrepancy regimes for the Gaussian normalized power prior experiment.

5.2.1.1 Discussion of results

The results reveal a clear distinction between methods that directly target the exact posterior distribution and methods that rely on noisier approximations of the normalizing constant ratio.

First, the Exchange MH algorithm is nearly indistinguishable from the analytic reference posterior across all discrepancy regimes. This confirms empirically that the exchange construction correctly removes the intractable normalizing constant ratio while preserving the target posterior distribution. Moreover, because auxiliary draws from the powered prior are available analytically in the Gaussian setting, the exchange update remains computationally efficient despite being exact.

Scenario	Method	Batches	Time (s)	W_η	$W_{\|\theta\|}$	ESS_η	$\text{ESS}_{\|\theta\|}$	Acc. η
Small	Analytic MH (exact reference)	5	100.6	0.000	0.000	552	2327	0.978
	Exchange MH	1	9.5	0.005	0.006	63	226	0.953
	Prior IS	10	2278.2	0.072	0.002	517	2111	0.965
	MLE-centered IS	7	4614.7	0.005	0.003	387	1421	0.978
	Path-sampling PMMH	6	5812.2	0.008	0.003	333	1234	0.978
Medium	Analytic MH (exact reference)	5	97.7	0.000	0.000	1046	2677	0.907
	Exchange MH	1	9.7	0.008	0.004	123	266	0.847
	Prior IS	1	221.5	0.006	0.007	175	362	0.909
	MLE-centered IS	4	2553.1	0.009	0.003	229	945	0.907
	Path-sampling PMMH	1	968.2	0.008	0.011	161	272	0.911
Large	Analytic MH (exact reference)	5	94.5	0.000	0.000	18361	3474	0.749
	Exchange MH	1	9.2	0.001	0.007	1474	401	0.448
	Prior IS	1	178.9	0.001	0.006	872	390	0.748
	MLE-centered IS	1	513.6	0.000	0.004	1923	350	0.748
	Path-sampling PMMH	1	775.7	0.000	0.004	1954	398	0.753

Table 1 – Experiment 1 convergence diagnostics. Batches: number of batches of 4 chains × 5000 iterations required to reach $W_1(\eta) < 0.01$. Wasserstein distances computed against the analytic MH reference.

Scenario	Method	$\bar{\eta}$	sd(η)	$\overline{\|\theta\|}$	sd($\|\theta\|$)
Small	Analytic MH (exact reference)	0.623	0.241	0.223	0.089
	Exchange MH	0.621	0.246	0.228	0.087
	Prior IS	0.695	0.238	0.222	0.089
	MLE-centered IS	0.626	0.244	0.225	0.090
	Path-sampling PMMH	0.615	0.243	0.226	0.088
Medium	Analytic MH (exact reference)	0.174	0.138	0.291	0.115
	Exchange MH	0.182	0.147	0.291	0.113
	Prior IS	0.173	0.128	0.287	0.110
	MLE-centered IS	0.183	0.160	0.291	0.118
	Path-sampling PMMH	0.169	0.124	0.296	0.105
Large	Analytic MH (exact reference)	0.031	0.024	0.268	0.107
	Exchange MH	0.032	0.025	0.264	0.103
	Prior IS	0.032	0.026	0.262	0.105
	MLE-centered IS	0.032	0.024	0.270	0.109
	Path-sampling PMMH	0.031	0.024	0.265	0.108

Table 2 – Experiment 1 posterior summaries (Normal-Normal NPP).

The Prior IS and MLE-centered IS methods both exhibit sensitivity to the discrepancy regime. Under low discrepancy, where the posterior distribution of η is diffuse, the variance of the importance weights becomes more pronounced and larger computational effort is required to achieve the target Wasserstein tolerance. The MLE-centered proposal generally improves stability relative to Prior IS by concentrating importance samples near regions of higher likelihood support.

As the discrepancy increases, the posterior distribution of η concentrates near zero. In this regime, all methods become substantially easier to stabilize because the relevant range of powered likelihoods becomes smaller. Consequently, the Wasserstein distances decrease rapidly and the differences between methods become less pronounced.

The Path Sampling PMMH method displays the largest variability across discrepancy scenarios. Because the thermodynamic integration estimator depends on Monte Carlo approximations along an entire temperature path, estimator variability accumulates and may degrade mixing efficiency. This effect is especially visible in diffuse posterior regimes.

Overall, the experiments demonstrate that exact inference for the normalized power prior is computationally feasible through the Exchange algorithm, even in settings where direct evaluation of the normalizing constant ratio would otherwise be computationally prohibitive. The exchange sampler additionally provides a principled exact baseline against which approximate pseudo-marginal and future surrogate-based methods may be assessed.

5.2.2 Linear regression with Normal-Inverse-Gamma priors

We next consider a Gaussian linear regression model with unknown variance. This experiment is designed to evaluate the exchange update in a setting where the normalized power prior remains analytically tractable, but the normalizing constant is no longer a simple linear function of the power parameter. Let $D_0 = (\boldsymbol{X}_0, \boldsymbol{y}_0)$ denote the historical data and let $D = (\boldsymbol{X}, \boldsymbol{y})$ denote the current data. We assume

$$\boldsymbol{y}_0 = \boldsymbol{X}_0\boldsymbol{\beta} + \boldsymbol{\varepsilon}_0, \qquad \boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{5.4}$$

where $\boldsymbol{\varepsilon}_0 \sim \mathcal{N}(\mathbf{0}, \sigma^2 \boldsymbol{I}_{n_0})$ and $\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma^2 \boldsymbol{I}_n)$. The variance σ^2 is unknown. The initial prior is Normal-Inverse-Gamma,

$$\boldsymbol{\beta} \mid \sigma^2 \sim \mathcal{N}(\boldsymbol{\mu}_0, \sigma^2 \boldsymbol{V}_0), \qquad \sigma^2 \sim \text{Inv-Gamma}(a, b). \tag{5.5}$$

For a fixed value of η, the powered prior is given by

$$\pi_\eta(\boldsymbol{\beta}, \sigma^2 \mid D_0) = \frac{L(D_0 \mid \boldsymbol{\beta}, \sigma^2)^\eta \pi_0(\boldsymbol{\beta}, \sigma^2)}{c_0(\eta)}. \tag{5.6}$$

The normalizing constant is

$$c_0(\eta) = \int_0^\infty \int_{\mathbb{R}^p} L(D_0 \mid \boldsymbol{u}, v)^\eta \pi_0(\boldsymbol{u}, v) \mathrm{d}\boldsymbol{u} \mathrm{d}v. \tag{5.7}$$

In this model, $c_0(\eta)$ is available in closed form by conjugacy. This allows us to construct an analytic reference chain for η and to validate algorithms that do not directly evaluate $c_0(\eta)$. The marginal posterior density of η is proportional to

$$\pi(\eta \mid D_0, D) \propto \frac{\pi_A(\eta)\Gamma(n/2 + \nu_0)|\Lambda_0|^{1/2} H_0(\eta)^{\nu_0}}{|\Lambda|^{1/2}\Gamma(\nu_0) H(\eta)^{\nu_0 + n/2}}. \tag{5.8}$$

The quantities ν_0, Λ_0, $H_0(\eta)$, Λ, and $H(\eta)$ are defined in Appendix B.2. Although the analytic expression is available here, the exchange algorithm does not require evaluating $c_0(\eta)$. Given the current state $(\boldsymbol{\beta}, \sigma^2, \eta)$, we propose η' and draw an auxiliary value

$$(\boldsymbol{\beta}^{\star}, \sigma^{2\star}) \sim \pi_{\eta'}(\boldsymbol{\beta}, \sigma^2 \mid D_0). \tag{5.9}$$

The exchange log-ratio for the η update is

$$\begin{aligned}\log R_{\text{ex}} = (\eta' - \eta) \log L(D_0 \mid \boldsymbol{\beta}, \sigma^2) + (\eta - \eta') \log L(D_0 \mid \boldsymbol{\beta}^{\star}, \sigma^{2\star}) \\ + \log \frac{\pi_A(\eta')}{\pi_A(\eta)} + \log \frac{q(\eta \mid \eta')}{q(\eta' \mid \eta)}.\end{aligned} \tag{5.10}$$

The proposal is then accepted with probability $\min\{1, \exp(\log R_{\text{ex}})\}$. When the auxiliary draw from $\pi_{\eta'}(\boldsymbol{\beta}, \sigma^2 \mid D_0)$ is exact, this update has the correct invariant distribution for the normalized power prior. This follows from the exchange algorithm for doubly-intractable distributions, where an auxiliary draw from the proposed normalized model cancels the unknown ratio of normalizing constants in the Metropolis-Hastings acceptance probability (Murray; Ghahramani; MacKay, 2012). The exchange algorithm therefore provides an exact update for η under the powered-prior simulation assumption. This assumption is satisfied in the present experiment because the powered prior is Normal-Inverse-Gamma.

We compare three methods. The first method is an analytic Metropolis-Hastings reference that evaluates Equation (5.8). The second method is the exchange Metropolis-Hastings update in Equation (5.10). The third method is a path-sampling pseudo-marginal method based on a Monte Carlo approximation of the normalizing constant ratio. The path-sampling method is included as a computational baseline and is classified according to the exactness of the estimator used in its acceptance ratio. If the estimator is not non-negative and unbiased for the required marginal target, the method is treated as approximate rather than exact. This distinction follows the pseudo-marginal principle, where exactness requires a non-negative unbiased estimator of the target density on an extended space (Andrieu; Roberts, 2009).

We generate three discrepancy scenarios between the historical and current regression coefficients. In the small-discrepancy scenario, the current coefficient vector is close to the historical coefficient vector. In the medium-discrepancy scenario, the current coefficient vector is moderately shifted. In the large-discrepancy scenario, the current coefficient vector is substantially shifted. The expected qualitative behavior is that the posterior distribution of η moves toward smaller values as the discrepancy increases. This reflects reduced borrowing from the historical data when the historical and current data-generating mechanisms are less compatible.

The main diagnostic is the one-dimensional Wasserstein distance from each method to the analytic reference. We report this distance separately for η, $\|\boldsymbol{\beta}\|_2$, and σ^2. We also report posterior means, posterior standard deviations, effective sample sizes, acceptance rates, and runtime. The norm $\|\boldsymbol{\beta}\|_2$ summarizes the magnitude of the regression coefficient vector, while σ^2 is reported separately because it represents the residual variance and should not be combined with $\boldsymbol{\beta}$ into a single geometric average.

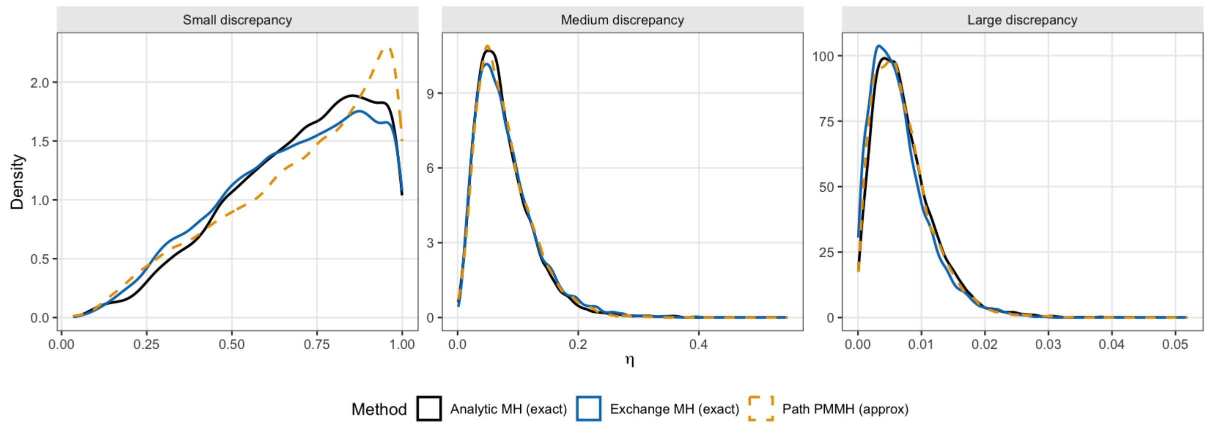

Figure 3 – Posterior density of η for the linear regression NPP experiment across discrepancy scenarios.

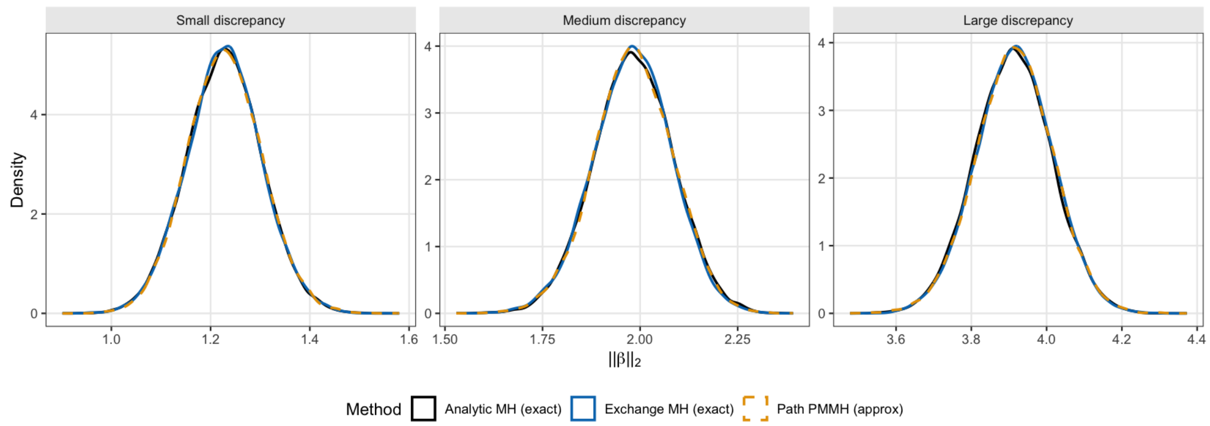

Figure 4 – Posterior density of $\|\boldsymbol{\beta}\|_2$ for the linear regression NPP experiment across discrepancy scenarios.

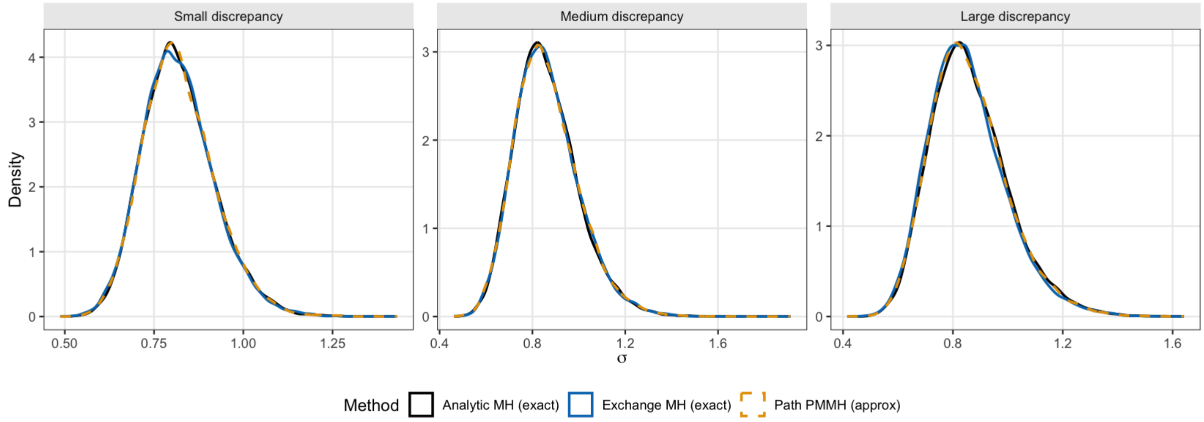

Figure 5 – Posterior density of σ^2 for the linear regression NPP experiment across discrepancy scenarios.

Scenario	Method	W_η	$W_{\lVert\beta\rVert}$	W_σ	ESS_η	Acc. η	Time (s)
Small	Analytic MH	0.000	0.000	0.000	195	0.928	3.2
Small	Exchange MH	0.019	0.001	0.002	222	0.902	1.6
Small	Path PMMH	0.022	0.001	0.001	199	0.913	34.3
Medium	Analytic MH	0.000	0.000	0.000	717	0.859	3.2
Medium	Exchange MH	0.003	0.003	0.002	557	0.797	1.7
Medium	Path PMMH	0.002	0.001	0.002	797	0.837	34.8
Large	Analytic MH	0.000	0.000	0.000	682	0.863	3.3
Large	Exchange MH	0.001	0.003	0.009	577	0.824	1.8
Large	Path PMMH	0.000	0.003	0.003	694	0.858	36.7

Table 3 – Diagnostic comparison for the linear regression NPP experiment with Normal-Inverse-Gamma priors. Wasserstein distances are computed against the analytic MH baseline within each discrepancy scenario.

Scenario	Method	$\bar{\eta}$	sd(η)	$\overline{\lVert\beta\rVert}$	sd($\lVert\beta\rVert$)	$\bar{\sigma}$	sd(σ)
Small	Analytic MH	0.697	0.208	1.227	0.076	0.820	0.101
Small	Exchange MH	0.679	0.217	1.228	0.075	0.818	0.101
Small	Path PMMH	0.705	0.227	1.229	0.075	0.821	0.101
Medium	Analytic MH	0.077	0.049	1.984	0.102	0.862	0.137
Medium	Exchange MH	0.080	0.051	1.981	0.101	0.863	0.137
Medium	Path PMMH	0.078	0.047	1.983	0.102	0.863	0.138
Large	Analytic MH	0.007	0.005	3.912	0.102	0.863	0.142
Large	Exchange MH	0.007	0.005	3.915	0.101	0.854	0.140
Large	Path PMMH	0.007	0.005	3.915	0.103	0.860	0.141

Table 4 – Posterior summaries for the NIG linear regression NPP experiment.

5.2.3 Logistic regression with surrogate normalizing constants

This experiment considers a logistic regression model for which the normalizing constant of the normalized power prior is not available in closed form. The objective is to evaluate whether surrogate models can accurately emulate the function $c_0(\eta)$ and thereby provide a computationally efficient alternative to exact inference.

Let $D_0 = (\boldsymbol{X}_0, \boldsymbol{y}_0)$ denote the historical dataset and $D = (\boldsymbol{X}, \boldsymbol{y})$ denote the current dataset. We assume

$$y_{0i} \mid \boldsymbol{\beta} \sim \text{Bernoulli}(p_{0i}), \qquad \text{logit}(p_{0i}) = \boldsymbol{x}_{0i}^{\top}\boldsymbol{\beta}, \tag{5.11}$$

and

$$y_i \mid \boldsymbol{\beta} \sim \text{Bernoulli}(p_i), \qquad \text{logit}(p_i) = \boldsymbol{x}_i^{\top}\boldsymbol{\beta}. \tag{5.12}$$

The regression coefficient vector has prior distribution $\boldsymbol{\beta} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{I}_p)$, with $p = 3$.

Unlike the Gaussian and Normal–Inverse–Gamma examples considered previously, no closed-form expression is available for $c_0(\eta)$. Consequently, direct Metropolis–Hastings updates involving the ratio $c_0(\eta')/c_0(\eta)$ are not feasible.

To obtain an exact reference posterior, we employ the Exchange algorithm. For a proposed value η', an auxiliary draw

$$\boldsymbol{\beta}^{\star} \sim \pi_{\eta'}(\boldsymbol{\beta} \mid D_0) \tag{5.13}$$

is generated using Stan. The resulting exchange acceptance ratio depends only on likelihood evaluations and does not require explicit computation of the normalizing constant ratio. Under exact simulation from the powered prior, the resulting Markov chain targets the correct normalized power prior posterior distribution.

The remaining methods replace $c_0(\eta)$ by an emulator constructed from a finite set of training evaluations.

We consider three surrogate models. The first is a spline emulator based on smoothing splines fitted to estimates of $\log c_0(\eta)$ over a grid of power parameters. The second is a Gaussian process emulator, which treats $\log c_0(\eta)$ as an unknown latent function and uses a probabilistic interpolation model to predict its value at unobserved values of η. The third is a shape-constrained additive model (SCAM), which incorporates structural information about the normalizing constant through monotonicity constraints.

Since

$$\frac{\partial}{\partial \eta} \log c_0(\eta) = \mathbb{E}_{\pi\eta}\left[\log L(D_0 \mid \boldsymbol{\beta})\right], \tag{5.14}$$

the derivative process provides information about the qualitative shape of $\log c_0(\eta)$, which we incorporate through constrained smoothers, since the function $\eta \mapsto \log c_0(\eta)$

is convex, as shown in Appendix A.4. These structural properties motivate the use of shape-constrained emulators in Section 2.6.2.

In all cases, we emulate the log-normalizing constant using $K = 3000$ points sampled via importance sampling.

Three discrepancy scenarios are generated by increasing the separation between the historical and current regression coefficients. As in the previous experiments, increasing discrepancy is expected to shift posterior mass toward smaller values of η, indicating reduced borrowing from historical information.

The Exchange sampler is treated as the reference method. For each surrogate model, we report Wasserstein distances relative to the Exchange posterior, together with effective sample sizes, acceptance rates, posterior means, posterior standard deviations, and computational runtime.

Posterior agreement is assessed for η and for each regression coefficient separately. This allows distortions introduced by the surrogate approximation to be localized to specific components of the parameter vector.

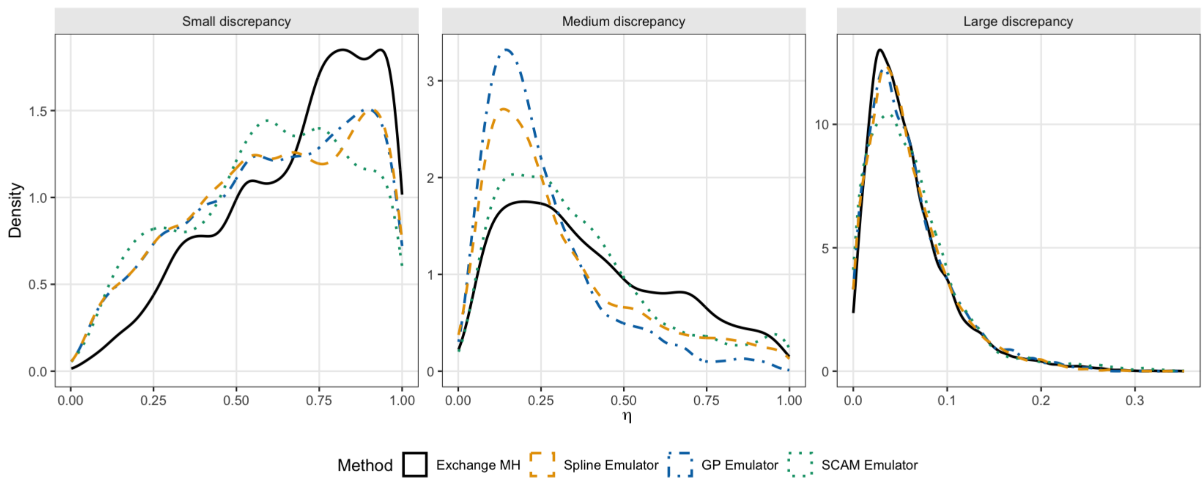

Figure 6 – Posterior density of η across discrepancy scenarios for the logistic regression normalized power prior experiment.

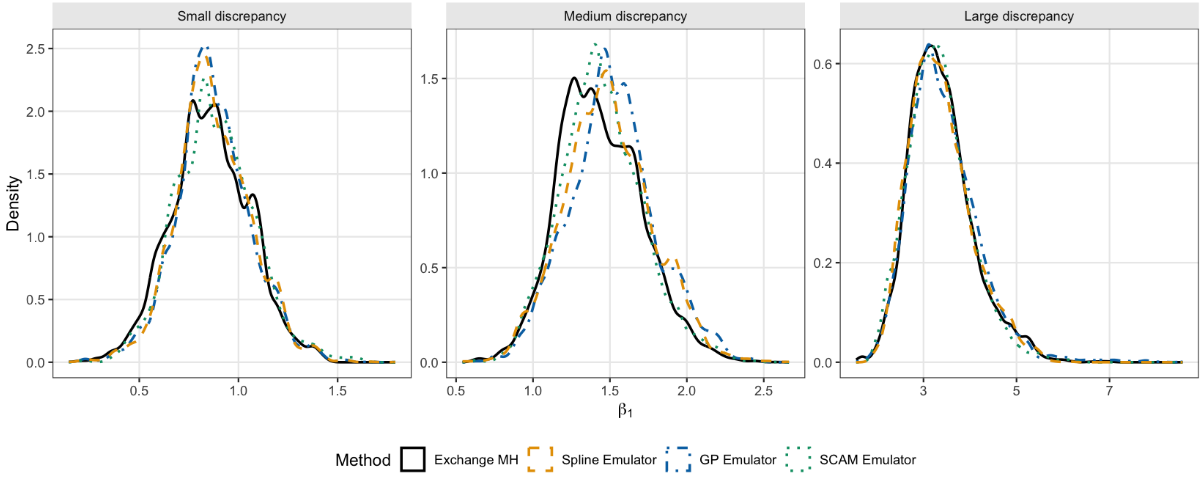

Figure 7 – Posterior density of the first regression coefficient across discrepancy scenarios.

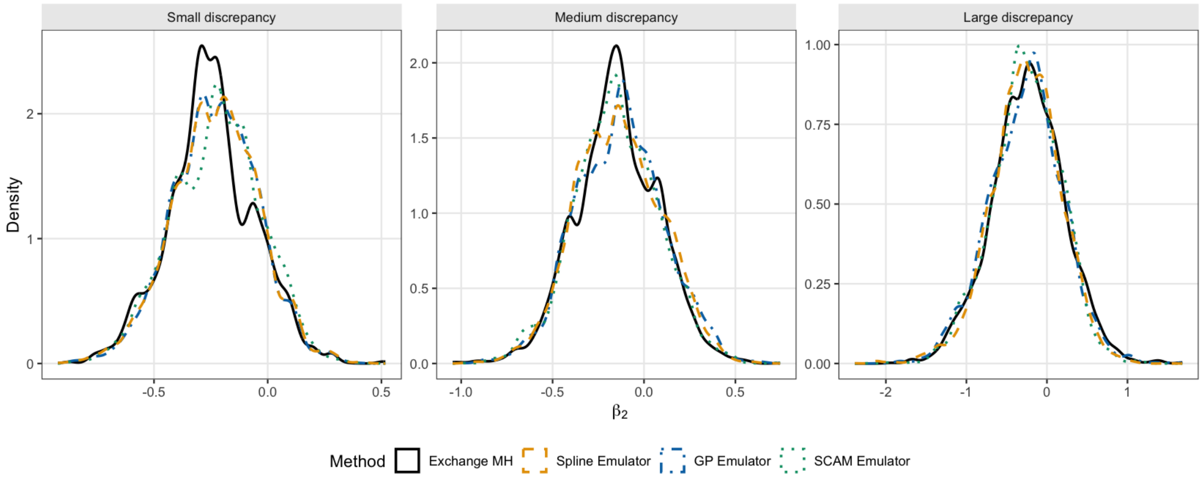

Figure 8 – Posterior density of the second regression coefficient across discrepancy scenarios.

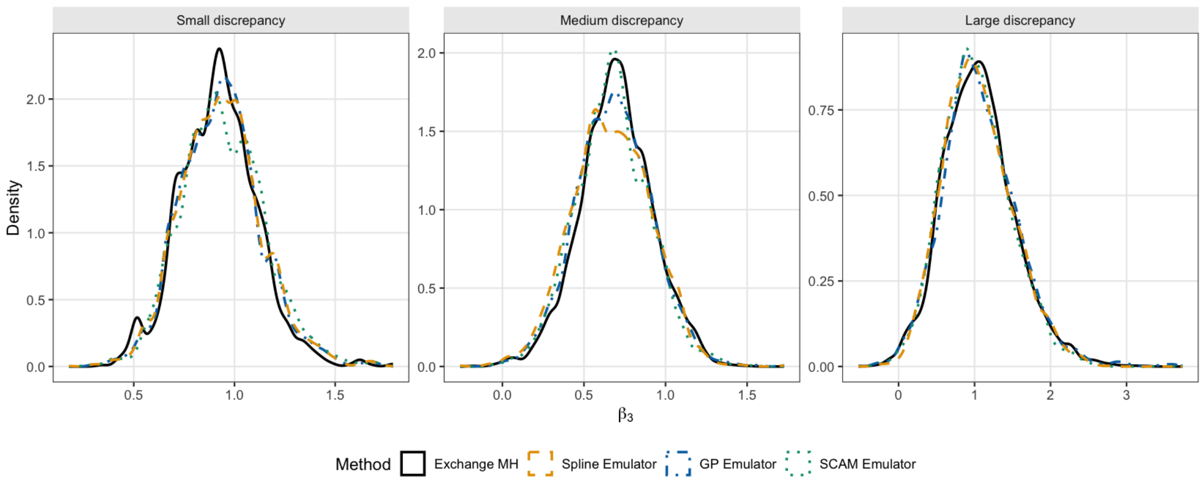

Figure 9 – Posterior density of the third regression coefficient across discrepancy scenarios.

Scenario	Method	W_η	W_{β_1}	W_{β_2}	W_{β_3}	ESS_η	Acc. η	Time (s)
Small	Exchange MH	0.000	0.000	0.000	0.000	56	0.947	2543.7
Small	Spline Emulator	0.074	0.017	0.018	0.015	33	0.955	1.1
Small	GP Emulator	0.070	0.019	0.017	0.013	33	0.956	6.5
Small	SCAM Emulator	0.091	0.023	0.023	0.025	30	0.957	15.2
Medium	Exchange MH	0.000	0.000	0.000	0.000	33	0.918	2619.5
Medium	Spline Emulator	0.087	0.052	0.016	0.029	26	0.921	0.9
Medium	GP Emulator	0.145	0.080	0.019	0.016	29	0.920	6.1
Medium	SCAM Emulator	0.041	0.015	0.010	0.025	41	0.933	15.7
Large	Exchange MH	0.000	0.000	0.000	0.000	309	0.604	2353.0
Large	Spline Emulator	0.002	0.024	0.030	0.021	411	0.700	0.9
Large	GP Emulator	0.002	0.061	0.028	0.020	397	0.682	5.8
Large	SCAM Emulator	0.005	0.044	0.024	0.038	252	0.715	14.1

Table 5 – Diagnostic comparison for the logistic regression NPP experiment. Wasserstein distances are computed against the exchange MH reference within each discrepancy scenario.

Scenario	Method	$\bar{\eta}$	sd(η)	$\bar{\beta}_1$	sd(β_1)	$\bar{\beta}_2$	sd(β_2)	$\bar{\beta}_3$	sd(β_3)
Small	Exchange MH	0.681	0.227	0.860	0.198	-0.246	0.190	0.922	0.192
Small	Spline Emulator	0.608	0.254	0.875	0.189	-0.229	0.191	0.934	0.201
Small	GP Emulator	0.611	0.254	0.874	0.187	-0.231	0.187	0.932	0.199
Small	SCAM Emulator	0.590	0.248	0.882	0.200	-0.224	0.195	0.944	0.207
Medium	Exchange MH	0.395	0.245	1.437	0.270	-0.146	0.225	0.698	0.222
Medium	Spline Emulator	0.308	0.229	1.489	0.284	-0.140	0.234	0.669	0.241
Medium	GP Emulator	0.250	0.177	1.517	0.281	-0.132	0.237	0.684	0.231
Medium	SCAM Emulator	0.355	0.230	1.448	0.267	-0.153	0.228	0.674	0.226
Large	Exchange MH	0.059	0.044	3.383	0.687	-0.235	0.452	1.062	0.473
Large	Spline Emulator	0.059	0.043	3.370	0.675	-0.261	0.432	1.047	0.455
Large	GP Emulator	0.060	0.044	3.444	0.753	-0.263	0.457	1.075	0.496
Large	SCAM Emulator	0.063	0.050	3.340	0.661	-0.254	0.425	1.026	0.448

Table 6 – Posterior summaries for the logistic regression NPP experiment across three discrepancy levels.

5.2.3.1 Discussion of results

`Adame:` Discuss here the results. For me, the most important is that the betas (that are the quantitiy of interest) do not change much even when the eta inteference does not match. Also, the Exact is way more slow, and we can only see a difference between them all when the discrepancy is medium, which suggests that the surrogate methods tend to concentrate on 0 more easily (splines and GP, compared to SCAM and Exchange).

The results reveal an important asymmetry between the inferential impact of approximation errors on η and on $\boldsymbol{\beta}$. Even in cases where the surrogate model introduces notable distortion in the marginal posterior of η, the posterior distributions of the regression coefficients $\boldsymbol{\beta}$ remain largely unaffected. This suggests that, for the quantity of primary inferential interest in regression applications, moderate errors in the approximation of

$c_0(\eta)$ may be practically tolerable.

A further notable finding is that all methods produce the most discernible differences under the medium-discrepancy scenario. Under large discrepancy, the posterior of η concentrates near zero regardless of the method, reducing the sensitivity to approximation errors.Under small discrepancy, all posteriors are diffuse and similar.

It is precisely the intermediate regime, where the posterior of η is neither degenerate nor flat, that discriminates most sharply between exact and approximate approaches. The spline and GP surrogates tend to concentrate posterior mass towards zero more aggressively than SCAM and Exchange in this regime, suggesting that they may underestimate the degree of historical borrowing when data are moderately compatible.

Finally, the Exchange sampler is considerably slower per iteration than the surrogate methods, reflecting the cost of exact auxiliary draws via Stan. This computational gap motivates the development of faster exact alternatives, such as the pseudo-marginal approach of Section 4.3, as a practical middle ground.

6 Discussion

Code Availability The code is available at: <https://github.com/adamesalles/npp-msc-thesis>.

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APPENDIX A – Proofs

A.1 Thermodynamic identity for the normalising constant

Proof. Under regularity conditions permitting differentiation under the integral sign, we have

$$\frac{\mathrm{d}c_0(\eta)}{\mathrm{d}\eta} = \frac{\mathrm{d}}{\mathrm{d}\eta}\int_\Theta L_0(D_0 \mid t)^\eta \,\mathrm{d}P_0(t) \tag{A.1}$$

$$= \int_\Theta \frac{\mathrm{d}}{\mathrm{d}\eta} L_0(D_0 \mid t)^\eta \,\mathrm{d}P_0(t) \tag{A.2}$$

$$= \int_\Theta \log L_0(D_0 \mid t) \cdot L_0(D_0 \mid t)^\eta \,\mathrm{d}P_0(t) \tag{A.3}$$

$$= c_0(\eta) \int_\Theta \log L_0(D_0 \mid t) \cdot \pi_\eta(t \mid D_0)\,\mathrm{d}t \tag{A.4}$$

$$= c_0(\eta)\,\mathbb{E}_{\pi_\eta}\Big[\log L_0(D_0 \mid \theta)\Big]. \tag{A.5}$$

Since $c_0(\eta) > 0$ whenever the initial prior is proper, dividing both sides by $c_0(\eta)$ yields (2.30). □

A.2 Unbiasedness of the Poisson estimator

Proof. We use the notation of Section 4.2.1: write $\lambda := \lambda_{\eta,\eta'}$, $c := c_{\eta,\eta'}$, and $\Delta := \Delta(\eta, \eta')$. Since $(U_j, \theta_j) \overset{\text{iid}}{\sim} \mathsf{P}_{\eta,\eta'}$ and $K \sim \text{Poisson}(\lambda)$ are drawn independently, the tower property gives

$$\mathbb{E}\Big[\widehat{I}(\eta,\eta')\Big] = \exp\{\lambda - c\}\,\mathbb{E}_K\left[\left(\mathbb{E}_{\mathsf{P}_{\eta,\eta'}}\left[\frac{c - Z_{\eta,\eta'}}{\lambda}\right]\right)^K\right] \tag{A.6}$$

$$= \exp\{\lambda - c\}\,\mathbb{E}_K\left[\left(\frac{c-\Delta}{\lambda}\right)^K\right]. \tag{A.7}$$

Applying the probability-generating function of a Poisson(λ) random variable, $\mathbb{E}[r^K] = \exp\{\lambda(r-1)\}$, with $r = (c - \Delta)/\lambda$, gives

$$\mathbb{E}\Big[\widehat{I}(\eta,\eta')\Big] = \exp\{\lambda - c\} \cdot \exp\left\{\lambda\left(\frac{c-\Delta}{\lambda} - 1\right)\right\} \tag{A.8}$$

$$= \exp\{\lambda - c\} \cdot \exp\{c - \Delta - \lambda\} \tag{A.9}$$

$$= \exp\{-\Delta\} = \frac{c_0(\eta)}{c_0(\eta')}, \tag{A.10}$$

which establishes the claim. □

A.3 Correctness of the exchange algorithm

Proof. We verify detailed balance for the augmented kernel. Let $\pi(\eta \mid \theta) \propto \pi_A(\eta)\, L_0(D_0 \mid \theta)^\eta / c_0(\eta)$ denote the conditional target for η. The exchange algorithm augments the state space with $\theta^\star$ drawn from the proposed normalized model:

$$\theta^\star \sim \pi_{\eta'}(\theta \mid D_0) = \frac{L_0(D_0 \mid \theta)^{\eta'}}{c_0(\eta')}\, \pi_0(\theta). \tag{A.11}$$

The augmented target on $(\eta, \theta^\star)$ is

$$\widetilde{\pi}(\eta, \theta^\star) \propto \pi(\eta \mid \theta) \cdot \pi_{\eta'}(\theta^\star \mid D_0). \tag{A.12}$$

The proposed move $(\eta, \theta^\star) \to (\eta', \theta)$ is accepted with the Metropolis–Hastings ratio

$$R_{\text{ex}} = \frac{\widetilde{\pi}(\eta', \theta)\, q(\eta \mid \eta')}{\widetilde{\pi}(\eta, \theta^\star)\, q(\eta' \mid \eta)} = \frac{\pi_A(\eta')\, L_0(D_0 \mid \theta)^{\eta'} / c_0(\eta')}{\pi_A(\eta)\, L_0(D_0 \mid \theta)^{\eta} / c_0(\eta)} \cdot \frac{L_0(D_0 \mid \theta)^{\eta} / c_0(\eta)}{L_0(D_0 \mid \theta^\star)^{\eta'} / c_0(\eta')} \cdot \frac{q(\eta \mid \eta')}{q(\eta' \mid \eta)}. \tag{A.13}$$

The factors $c_0(\eta)$ and $c_0(\eta')$ cancel, yielding

$$R_{\text{ex}} = \frac{\pi_A(\eta')}{\pi_A(\eta)} \cdot \frac{q(\eta \mid \eta')}{q(\eta' \mid \eta)} \cdot \frac{L_0(D_0 \mid \theta)^{\eta' - \eta}}{L_0(D_0 \mid \theta^\star)^{\eta' - \eta}}, \tag{A.14}$$

which is exactly the ratio in Section 4.1. Since the augmented kernel satisfies detailed balance with respect to $\widetilde{\pi}$, and since the marginal of $\widetilde{\pi}$ over $\theta^\star$ recovers $\pi(\eta \mid \theta)$, the exchange update has the correct invariant distribution. □

A.4 Convexity of $\log c_0(\eta)$

Proof. We show that $\eta \mapsto \log c_0(\eta)$ is convex on $[0, 1]$. By (2.30), the second derivative satisfies

$$\frac{\mathrm{d}^2 \log c_0(\eta)}{\mathrm{d}\eta^2} = \frac{\mathrm{d}}{\mathrm{d}\eta}\, \mathbb{E}_{\pi_\eta}\Big[\log L_0(D_0 \mid \theta)\Big] = \mathrm{Var}_{\pi_\eta}\Big[\log L_0(D_0 \mid \theta)\Big] \geq 0, \tag{A.15}$$

where the identity follows by differentiating under the integral sign and recognising the resulting expression as a variance under π_η. Since the second derivative is non-negative, $\log c_0(\eta)$ is convex. □

APPENDIX B – Analytic Formulations

This appendix collects closed-form expressions for the normalising constant $c_0(\eta)$ and its derivative under specific model families. These results are used in the experiments of Chapter 4 to construct analytic reference chains and to validate the unbiased estimators developed there.

B.1 Gaussian likelihood with known variance

Consider $d_{0j} \sim \mathcal{N}(\theta, \sigma^2)$ with known variance σ^2 for all $j = 1, \ldots, N_0$, and let P_0 be the measure induced by the prior $\pi_0(\theta) \propto 1$.

Therefore, let $\bar{d}_{0p} = \sum_{j=1}^{p} d_{0j}/p$ be the mean of the historical data D_0 and $S^2 = \sum_{j=1}^{p}(d_{0j} - \bar{d}_{0p})^2$ the sample variance. Then, we have

$$
\begin{aligned}
c_0(\eta) &= \int_{\Theta} L(D_0 \mid t)^{\eta} \, \mathrm{d}P_0(t), \\
&= \int_{-\infty}^{\infty} \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \exp\left\{ -\frac{\eta}{2\sigma^2} \left(\sum_{j=1}^{p} (d_{0j} - t)^2 \right) \right\} \mathrm{d}t, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \int_{-\infty}^{\infty} \exp\left\{ -\frac{\eta}{2\sigma^2} \left(\sum_{j=1}^{p} ((d_{0j} - \bar{d}_{0p}) + (\bar{d}_{0p} - t))^2 \right) \right\} \mathrm{d}t, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \int_{-\infty}^{\infty} \exp\left\{ -\frac{\eta}{2\sigma^2} \left(S^2 + p(t - \bar{d}_{0p})^2 + \cancelto{0}{\sum_{j=1}^{p} 2(d_{0j} - \bar{d}_{0p})(t - \bar{d}_{0p})} \right) \right\} \mathrm{d}t, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \exp\left\{ -\frac{\eta}{2\sigma^2} S^2 \right\} \int_{-\infty}^{\infty} \exp\left\{ -\frac{\eta}{2\sigma^2} p(t - \bar{d}_{0p})^2 \right\} \mathrm{d}t, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \exp\left\{ -\frac{\eta}{2\sigma^2} S^2 \right\} \sqrt{\frac{2\pi\sigma^2}{p\eta}} = \frac{1}{(2\pi\sigma^2)^{(\eta p - 1)/2} \sqrt{p\eta}} \exp\left\{ -\frac{\eta}{2\sigma^2} S^2 \right\}.
\end{aligned}
$$

Computing the logarithm of the normalizing constant, we have

$$
\log c_0(\eta) = -\frac{(\eta p - 1)}{2} \log\left(2\pi\sigma^2\right) - \frac{1}{2}\log(p\eta) - \frac{\eta}{2\sigma^2} S^2.
$$

Therefore, the derivative of the logarithm of the normalizing constant is given by

$$
\frac{\mathrm{d}\log c_0(\eta)}{\mathrm{d}\eta} = -\frac{p}{2}\log\left(2\pi\sigma^2\right) - \frac{S^2}{2\sigma^2} - \frac{1}{2\eta}.
$$

On the other hand, we can show that $\mathbb{E}_{\pi_\eta}[\log(L(D_0 \mid \theta))]$ matches the previous expression. In this case, we have that $\theta \mid D_0 \sim \mathcal{N}(\bar{d}_{0p}, \sigma^2/p\eta)$, so we can compute the expectation as follows:

$$
\begin{aligned}
\mathbb{E}_{\pi_\eta}[\log(L(D_0 \mid \theta))] &= -\frac{p}{2}\log\left(2\pi\sigma^2\right) - \mathbb{E}_{\pi_\eta}\left[\frac{1}{2\sigma^2}\left(\sum_{j=1}^{p}(d_{0j}-\theta)^2\right)\right], \\
&= -\frac{p}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{j=1}^{p}\mathbb{E}_{\pi_\eta}\left[(d_{0j}-\theta)^2\right], \\
&= -\frac{p}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{j=1}^{p}\left((d_{0j}-\mathbb{E}_{\pi_\eta}[\theta])^2 + \mathbb{V}_{\pi_\eta}[\theta]\right), \\
&= -\frac{p}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{j=1}^{p}\left((d_{0j}-\bar{d}_{0p})^2 + \frac{\sigma^2}{p\eta}\right), \\
&= -\frac{p}{2}\log\left(2\pi\sigma^2\right) - \frac{1}{2\sigma^2}\sum_{j=1}^{p}(d_{0j}-\bar{d}_{0p})^2 - \frac{p}{2\sigma^2}\frac{\sigma^2}{p\eta}, \\
&= -\frac{p}{2}\log\left(2\pi\sigma^2\right) - \frac{S^2}{2\sigma^2} - \frac{1}{2\eta}.
\end{aligned}
$$

B.1.1 Adding a Gaussian Initial Prior

Consider again $d_{0j} \sim \mathcal{N}(\theta, \sigma^2)$ with known variance σ^2 for all $j \in 0, \ldots, p$, but now let $P_0 \equiv \mathcal{N}(0, \sigma_0^2)$, where σ_0^2 is a known initial prior variance.

Again, let $\bar{d}_{0p} = \sum_{j=1}^{p} d_{0j}/p$ be the mean of the historical data D_0 and $S^2 = \sum_{j=1}^{p}(d_{0j} - \bar{d}_{0p})^2$ the sample variance. Starting, we get:

$$
c_0(\eta) = \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \exp\left\{-\frac{\eta}{2\sigma^2}S^2\right\} \frac{1}{\sqrt{2\pi\sigma_0^2}} \int_{-\infty}^{\infty} \exp\left\{-\frac{\eta}{2\sigma^2}p(t-\bar{d}_{0p})^2 - \frac{1}{2\sigma_0^2}t^2\right\} \mathrm{d}t.
$$

Expanding $-\frac{\eta p}{2\sigma^2}(\bar{d}_{0p} - t)^2 - \frac{t^2}{2\sigma_0^2}$, we get $(\bar{d}_{0p} - t)^2 = t^2 - 2t\bar{d}_{0p} + \bar{d}_{0p}^{\,2}$. And, thus, we have that

$$
\begin{aligned}
-\frac{\eta p}{2\sigma^2}(\bar{d}_{0p}-t)^2 - \frac{t^2}{2\sigma_0^2} &= -\frac{\eta p}{2\sigma^2}\left(t^2 - 2t\bar{d}_{0p} + \bar{d}_{0p}^{\,2}\right) - \frac{t^2}{2\sigma_0^2}, \\
&= -\frac{1}{2}\left(t^2\left(\frac{\eta p}{\sigma^2} + \frac{1}{\sigma_0^2}\right) - 2t\bar{d}_{0p}\frac{\eta p}{\sigma^2} + \bar{d}_{0p}^{\,2}\frac{\eta p}{\sigma^2}\right).
\end{aligned}
$$

Define $A = \frac{\eta p}{\sigma^2} + \frac{1}{\sigma_0^2}$ and $B = \frac{\eta p}{\sigma^2}\bar{d}_{0p}$. We can now complete the square. Once we have that $At^2 - 2Bt = A\left(t - \frac{B}{A}\right)^2 - \frac{B^2}{A}$, it follows that $-\frac{1}{2}(At^2 - 2Bt) = -\frac{A}{2}\left(t - \frac{B}{A}\right)^2 + \frac{B^2}{2A}$. Which gives us

$$
-\frac{1}{2}\left(At^2 - 2Bt + \bar{d}_{0p}^{\,2}\frac{\eta p}{\sigma^2}\right) = -\frac{A}{2}\left(t - \frac{B}{A}\right)^2 + \frac{B^2}{2A} - \frac{\bar{d}_{0p}^{\,2}}{2}\frac{\eta p}{\sigma^2}.
$$

Therefore, we can rewrite the normalizing constant as

$$
\begin{aligned}
c_0(\eta) &= \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \exp\left\{-\frac{\eta}{2\sigma^2}S^2\right\} \frac{1}{\sqrt{2\pi\sigma_0^2}} \int_{-\infty}^{\infty} \exp\left\{-\frac{A}{2}\left(t-\frac{B}{A}\right)^2 + \frac{B^2}{2A} - \frac{\bar{d}_{0p}^{\,2}}{2}\frac{\eta p}{\sigma^2}\right\} \mathrm{d}t, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \exp\left\{-\frac{\eta}{2\sigma^2}S^2 - \frac{\eta p}{2\sigma^2}\bar{d}_{0p}^{\,2}\right\} \cdot \frac{1}{\sqrt{A\sigma_0^2}} \cdot \exp\left\{\frac{B^2}{2A}\right\}, \\
&= \frac{1}{(2\pi\sigma^2)^{\eta p/2}} \exp\left\{-\frac{\eta}{2\sigma^2}\left(S^2 - p\bar{d}_{0p}^{\,2}\right) + \frac{B^2}{2A}\right\} \cdot \frac{1}{\sqrt{A\sigma_0^2}}.
\end{aligned}
$$

Taking the logarithm, we have

$$
\log c_0(\eta) = -\frac{(\eta p)}{2}\log\left(2\pi\sigma^2\right) - \frac{\eta}{2\sigma^2}\left(S^2 - p\bar{d}_{0p}^{\,2}\right) + \frac{B^2}{2A} - \frac{1}{2}\log\left(A\sigma_0^2\right).
$$

And, thus, we can compute the derivative of the logarithm of the normalizing constant term by term. The first two terms and the last one can be calculated right away:

$$
\begin{aligned}
\frac{\mathrm{d}}{\mathrm{d}\eta}\left(-\frac{(\eta p)}{2}\log\left(2\pi\sigma^2\right)\right) &= -\frac{p}{2}\log\left(2\pi\sigma^2\right), \\
\frac{\mathrm{d}}{\mathrm{d}\eta}\left(-\frac{\eta}{2\sigma^2}\left(S^2 - p\bar{d}_{0p}^{\,2}\right)\right) &= -\frac{1}{2\sigma^2}\left(S^2 - p\bar{d}_{0p}^{\,2}\right), \text{ and} \\
\frac{\mathrm{d}}{\mathrm{d}\eta}\left(-\frac{1}{2}\log\left(A\sigma_0^2\right)\right) &= -\frac{1}{2}\left(\frac{A'}{A}\right) = \frac{-p/\sigma^2}{2A}.
\end{aligned}
$$

The third term is given by

$$
\frac{d}{d\eta}\left(\frac{B^2}{2A}\right) = \frac{2BB'A - B^2A'}{2A^2},
$$

where $B' = \frac{d}{d\eta}\left(\frac{\eta p}{\sigma^2}\bar{d}_{0p}\right) = \frac{p}{\sigma^2}\bar{d}_{0p}$ and $A' = \frac{d}{d\eta}\left(\frac{\eta p}{\sigma^2} + \frac{1}{\sigma_0^2}\right) = \frac{p}{\sigma^2}$, thus

$$
\begin{aligned}
2BB' &= 2\cdot\left(\frac{\eta p}{\sigma^2}\bar{d}_{0p}\right)\cdot\left(\frac{p}{\sigma^2}\bar{d}_{0p}\right) = 2\eta\frac{p^2}{\sigma^4}\bar{d}_{0p}^{\,2}, \\
B^2 &= \left(\frac{\eta p}{\sigma^2}\bar{d}_{0p}\right)^2 = \eta^2\frac{p^2}{\sigma^4}\bar{d}_{0p}^{\,2},
\end{aligned}
$$

then

$$\frac{2BB'A - B^2A'}{2A^2} = \frac{1}{2A^2}\left(2\eta\frac{p^2}{\sigma^4}\bar{d}_{0p}^2 A - \eta^2\frac{p^3}{\sigma^6}\bar{d}_{0p}^2\right) = \frac{\eta\frac{p^2}{\sigma^4}\bar{d}_{0p}^2}{A^2}\left(A - \frac{\eta p}{2\sigma^2}\right).$$

And now we can recover the $c_0'(\eta)$ expression as

$$c_0'(\eta) = c_0(\eta)(\log(c_0(\eta)))' \tag{B.1}$$

and compare it with the importance sampling reformulation.

B.2 Linear Regression with NIG priors

Suppose we observe $\boldsymbol{y}_0 = \boldsymbol{X}_0\boldsymbol{\beta} + \varepsilon_0,\ \boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \varepsilon$ with $\varepsilon_0 \sim \mathcal{N}(0, \sigma^2 I_{n_0}),\ \varepsilon \sim \mathcal{N}(0, \sigma^2 I_n)$ and unknown variance σ^2. Now let $P_0 \equiv \mathcal{N}(\boldsymbol{\mu}_0,\ \boldsymbol{V}_0) \cdot \text{Inv-Gamma}(a, b)$, where $\boldsymbol{\mu}_0, \boldsymbol{V}_0, a$ and b are known hyperparameters.

Assume that $\boldsymbol{X}^\top\boldsymbol{X}$ is positive definite and let

$$\begin{aligned}
\hat{\boldsymbol{\beta}}_0 &= (\boldsymbol{X}_0^\top\boldsymbol{X}_0)^{-1}\boldsymbol{X}_0^\top\boldsymbol{y}_0,\\
\hat{\boldsymbol{\beta}} &= (\boldsymbol{X}^\top\boldsymbol{X})^{-1}\boldsymbol{X}^\top\boldsymbol{y},\\
S_0 &= (\boldsymbol{y}_0 - \boldsymbol{X}_0\hat{\boldsymbol{\beta}}_0)^\top(\boldsymbol{y}_0 - \boldsymbol{X}_0\hat{\boldsymbol{\beta}}_0),\\
S &= (\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}})^\top(\boldsymbol{y} - \boldsymbol{X}\hat{\boldsymbol{\beta}}).
\end{aligned}$$

So the likelihood of the historical data D_0 is

$$L(D_0 \mid \boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-n_0/2}\exp\left\{-\frac{1}{2\sigma^2}\left[S_0 + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_0)^\top\boldsymbol{X}_0^\top\boldsymbol{X}_0(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_0)\right]\right\}$$

and the initial prior of $(\boldsymbol{\beta}, \sigma^2)$ is

$$\pi_0(\boldsymbol{\beta}, \sigma^2) \propto (\sigma^2)^{-a-p/2}\exp\left\{-\frac{1}{\sigma^2}\left[b + \frac{1}{2}(\boldsymbol{\beta} - \boldsymbol{\mu}_0)^\top\boldsymbol{V}_0^{-1}(\boldsymbol{\beta} - \boldsymbol{\mu}_0)\right]\right\}.$$

Then

$$\begin{aligned}
c_0(\eta) &= \int_0^\infty\int_{\mathbb{R}^p}\pi_0(\boldsymbol{u}, v)L(D_0 \mid \boldsymbol{u}, v)^\eta d\boldsymbol{u}dv,\\
&\propto \int_0^\infty\int_{\mathbb{R}^p}(2\pi)^{-n_0\eta/2}v^{-a-p/2-n_0\eta/2}\exp\left\{-\frac{1}{2v}\left[2H_0(\eta) + (\boldsymbol{u} - \tilde{\boldsymbol{\beta}}_0)^\top(\boldsymbol{V}_0^{-1} + \eta\boldsymbol{X}_0^\top\boldsymbol{X}_0)(\boldsymbol{u} - \tilde{\boldsymbol{\beta}}_0)\right]\right\} d\\
&\propto (2\pi)^{-n_0\eta/2}\Gamma(\nu_0) \mid \Lambda_0 \mid^{-1/2} H_0(\eta)^{-\nu_0},
\end{aligned}$$

where

$$
\begin{aligned}
\tilde{\boldsymbol{\beta}}_0 &= (\Lambda_0)^{-1}(\boldsymbol{V}_0^{-1}\boldsymbol{\mu}_0 + \boldsymbol{X}_0^\top \boldsymbol{y}_0 \eta),\\
\nu_0 &= \frac{n_0\eta}{2} + a - 1,\\
\Lambda_0 &= \boldsymbol{V}_0^{-1} + \boldsymbol{X}_0^\top \boldsymbol{X}_0 \eta,\\
H_0(\eta) &= b + \frac{\eta}{2}\left[S_0 + (\boldsymbol{\mu}_0 - \hat{\boldsymbol{\beta}}_0)^\top \boldsymbol{X}_0^\top \boldsymbol{X}_0 (\Lambda_0)^{-1} \boldsymbol{V}_0^{-1}(\boldsymbol{\mu}_0 - \hat{\boldsymbol{\beta}}_0)\right].
\end{aligned}
$$

This leads to the following NPP:

$$
\pi(\boldsymbol{\beta}, \sigma^2, \eta \mid D_0) \propto \frac{\pi_0(\eta) H_0(\eta)^{\nu_0} \mid \Lambda_0 \mid^{1/2}}{(\sigma^2)^{\frac{n+n_0\eta+p}{2}+a}\Gamma(\nu_0)} \exp\left\{-\frac{1}{2\sigma^2}\left[(\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}_0)^\top \Lambda_0 (\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}_0) + 2H_0(\eta)\right]\right\}.
$$

Including the likelihood we can derive the joint posterior distribution:

$$
\pi(\boldsymbol{\beta}, \sigma^2, \eta \mid D, D_0) \propto \frac{\pi_0(\eta) H_0(\eta)^{\nu_0} \mid \Lambda_0 \mid^{1/2}}{(\sigma^2)^{\frac{n+n_0\eta+p}{2}+a}\Gamma(\nu_0)} \exp\left\{-\frac{1}{2\sigma^2}\left[(\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}})^\top \Lambda (\boldsymbol{\beta} - \tilde{\boldsymbol{\beta}}) + 2H(\eta)\right]\right\},
$$

where

$$
\begin{aligned}
\tilde{\boldsymbol{\beta}} &= \Lambda^{-1}(\boldsymbol{V}_0^{-1}\boldsymbol{\mu}_0 + \eta \boldsymbol{X}_0^\top \boldsymbol{y}_0 + \boldsymbol{X}^\top \boldsymbol{y}),\\
\Lambda &= \Lambda_0 + \boldsymbol{X}^\top \boldsymbol{X},\\
H(\eta) &= H_0(\eta) + \frac{1}{2}[S + (\tilde{\boldsymbol{\beta}}_0 - \hat{\boldsymbol{\beta}})^\top \boldsymbol{X}^\top \boldsymbol{X} \Lambda^{-1} \Lambda_0 (\tilde{\boldsymbol{\beta}}_0 - \hat{\boldsymbol{\beta}})].
\end{aligned}
$$

So we can express the marginal posterior distribution of η given (D_0, D) as

$$
\pi(\eta \mid D_0, D) \propto \frac{\pi_0(\eta)\Gamma(n/2 + \nu_0) \mid \Lambda_0 \mid^{1/2} H_0(\eta)^{\nu_0}}{\mid \Lambda \mid^{1/2} \Gamma(\nu_0) H(\eta)^{\nu_0 + n/2}},
$$

and the conditional posterior distribution of $\boldsymbol{\beta}$ given (η, D_0, D) follows the multivariate Student distribution with location parameter $\tilde{\boldsymbol{\beta}}$, shape matrix Σ, and degree of freedom ρ, where $\rho = n + 2\nu_0$ and $\Sigma = \frac{2H(\eta)}{n+2\nu_0}\Lambda^{-1}$.